

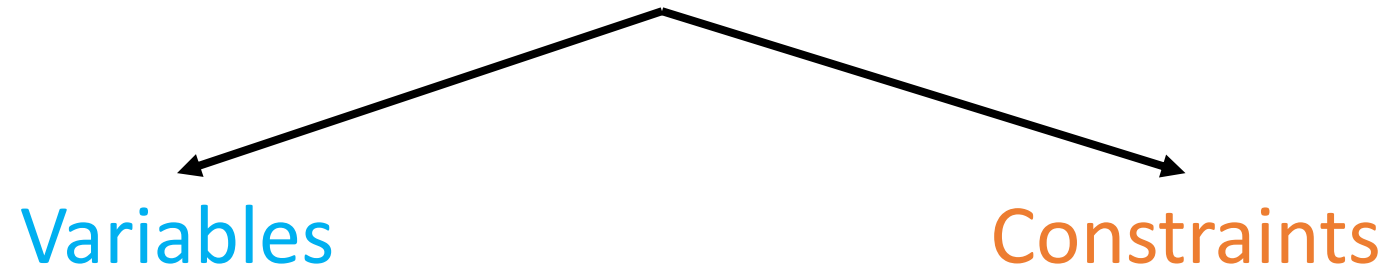
Part 1: Explanations and Checkers

ACP Summer School 2026

Emir Demirović



Modelling



```
graph TD; Modelling --> Variables; Modelling --> Constraints;
```

Variables

Constraints

Modelling

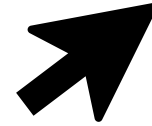
Variables

Constraints

$$X \in \mathcal{C} \subseteq \mathbb{N}^n$$

In this course:

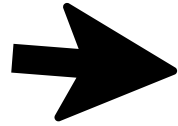
Constraint Programming



Constraints are arbitrary predicates

Constraints

$$X \in \mathcal{C} \subseteq \mathbb{N}^n$$



$$(x \vee y \vee z)$$

$$\sum w_i x_i \leq c$$

Constraints

$$X \in \mathcal{C} \subseteq \mathbb{N}^n$$

$$(x \vee y \vee z)$$

Expressive!



$$\sum w_i x_i \leq c$$

All-Different($x_1, x_2, \dots, x_n$)

NoOverlap($[x_{11}, x_{12}], [x_{21}, x_{22}], \dots, [x_{n1}, x_{n2}]$)

Proof-Theoretic Constraint Programming

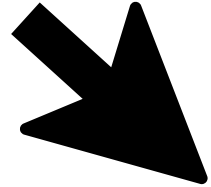
$$X \in \mathcal{C} \subseteq \mathbb{N}^n$$

“Lazy Clause Generation”

Proof-Theoretic Constraint Programming

$$X \in \mathcal{C} \subseteq \mathbb{N}^n$$

“Lazy Clause Generation”

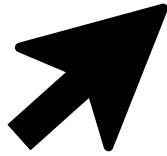


Combine constraints to derive new constraints

Proof-Theoretic Constraint Programming

$$X \in \mathcal{C} \subseteq \mathbb{N}^n$$

Part 1: Explanations and Checkers



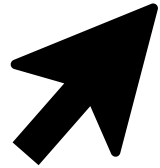
Correctness

Proof-Theoretic Constraint Programming

$$X \in \mathcal{C} \subseteq \mathbb{N}^n$$

Part 1: Explanations and Checkers

Part 2: Conflict Analysis and Proofs



Combine constraints

Proof-Theoretic Constraint Programming

$$X \in \mathcal{C} \subseteq \mathbb{N}^n$$

Part 1: Explanations and Checkers

Part 2: Conflict Analysis and Proofs



In class

Intuition

Visuals and Examples

Interaction!

Proof-Theoretic Constraint Programming

$$X \in \mathcal{C} \subseteq \mathbb{N}^n$$

Part 1: Explanations and Checkers

Part 2: Conflict Analysis and Proofs

In class

Intuition

Visuals and Examples

Interaction!

Out of class

Textbook (draft)
for details



Problem



Model

Problem

concrete

e.g., timetabling



Model

abstract

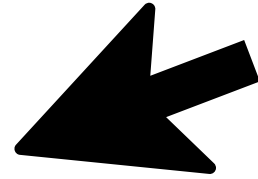
e.g., equations

unclear

Problem

concrete

e.g., timetabling



precise

Model

abstract

e.g., equations

unclear

Problem

concrete

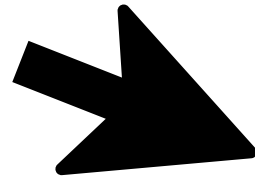
e.g., timetabling

precise

Model

abstract

e.g., equations



Solve

algorithms

unclear

Problem

concrete

e.g., timetabling

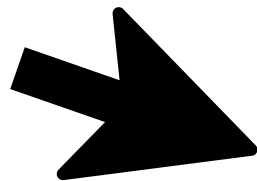
precise

Model

abstract

e.g., equations

Our focus



Solve

algorithms

$$y_1 + 2y_2 \geq 4$$

$$y_1 \in \{0, 1, 2, 3, 4\}$$

$$y_2 \in \{0, 2\}$$

Constraint

$$y_1 + 2y_2 \geq 4$$

Variables

$$y_1 \in \{0, 1, 2, 3, 4\}$$

$$y_2 \in \{0, 2\}$$

Constraint

$$y_1 + 2y_2 \geq 4$$

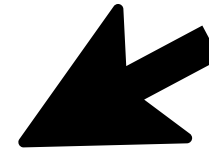
Variables

$$y_1 \in \{0, 1, 2, 3, 4\}$$

$$y_2 \in \{0, 2\}$$

Propagation

**Shrink domains
based on constraint**



Constraint

$$y_1 + 2y_2 \geq 4$$

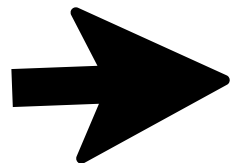
Variables

$$y_1 \in \{0, 1, 2, 3, 4\}$$

$$y_2 \in \{0, 2\}$$

Propagation

**Shrink domains
based on constraint**



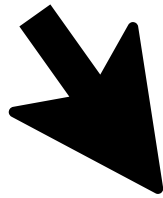
$$f: D^n \rightarrow D^n$$

$$y_1 + 2y_2 \geq 4$$

$$y_1 \in \{0, 1, 2, 3, 4\}$$

$$y_2 \in \{0, 2\}$$

Propagator



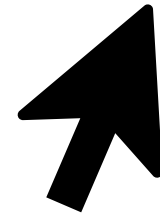
$$f(D_1, D_2) = f(\{0, 1, 2, 3, 4\}, \{0, 2\})$$

$$y_1 + 2y_2 \geq 4$$

$$y_1 \in \{0, 1, 2, 3, 4\}$$

$$y_2 \in \{0, 2\}$$

$$f(D_1, D_2) = f(\{0, 1, 2, 3, 4\}, \{0, 2\}) = (\{0, 1, 2, 3, 4\}, \{0, 2\})$$



No change: fixpoint

$$y_1 + 2y_2 \geq 4$$

$$y_1 \in \{0, 1, 2, 3, 4\}$$

$$y_2 \in \{0, 2\}$$

$$f(D_1, D_2) = f(\{0, 1, 2, 3, 4\}, \{0, 2\}) = (\{0, 1, 2, 3, 4\}, \{0, 2\})$$

$$f(D_1, D_2) = f(\{0, 1, 2, 3, 4\}, \{0\}) = \quad ?$$



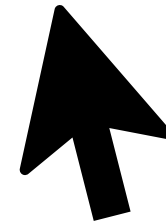
$$y_1 + 2y_2 \geq 4$$

$$y_1 \in \{0, 1, 2, 3, 4\}$$

$$y_2 \in \{0, 2\}$$

$$f(D_1, D_2) = f(\{0, 1, 2, 3, 4\}, \{0, 2\}) = (\{0, 1, 2, 3, 4\}, \{0, 2\})$$

$$f(D_1, D_2) = f(\{0, 1, 2, 3, 4\}, \{0\}) = (\{4\}, \{0\})$$



Domain changed!

$$y_1 + 2y_2 \geq 4$$

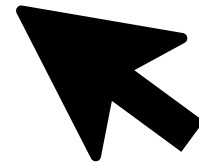
$$y_1 \in \{0, 1, 2, 3, 4\}$$

$$y_2 \in \{0, 2\}$$

$$f(D_1, D_2) = f(\{0, 1, 2, 3, 4\}, \{0, 2\}) = (\{0, 1, 2, 3, 4\}, \{0, 2\})$$

$$f(D_1, D_2) = f(\{0, 1, 2, 3, 4\}, \{0\}) = (\{4\}, \{0\})$$

$$f(D_1, D_2) = f(\{0, 1, 2, 3, \cancel{4}\}, \{0\}) = \quad ?$$



$$y_1 + 2y_2 \geq 4$$

$$y_1 \in \{0, 1, 2, 3, 4\}$$

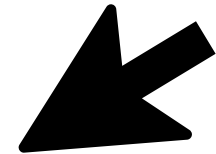
$$y_2 \in \{0, 2\}$$

$$f(D_1, D_2) = f(\{0, 1, 2, 3, 4\}, \{0, 2\}) = (\{0, 1, 2, 3, 4\}, \{0, 2\})$$

$$f(D_1, D_2) = f(\{0, 1, 2, 3, 4\}, \{0\}) = (\{4\}, \{0\})$$

$$f(D_1, D_2) = f(\{0, 1, 2, 3, \cancel{4}\}, \{0\}) = (\{\}, \{\})$$

Infeasible, conflict!



More complex example

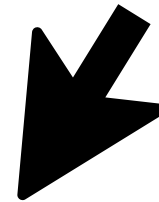
$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$



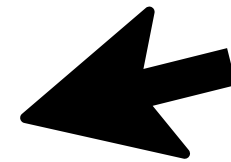
Propagate: remove values from domains!

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{\cancel{0}, 1, 4\}$$

$$y_2 \in \{\cancel{0}, 2\}$$

$$y_3 \in \{\cancel{0}, \cancel{1}, \cancel{2}, 3, 4\}$$



How to propagate?

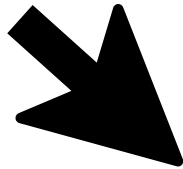
$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

“Upper Bound”



$$y_2 = \textcolor{blue}{UB}(y_2) = 2$$

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$y_2 = UB(y_2) = 2$$

$$y_3 = UB(y_3) = ?$$



$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$y_2 = UB(y_2) = 2$$

$$y_3 = UB(y_3) = 4$$



$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{0, 1, 4\}$$

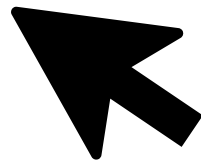
$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$y_2 = UB(y_2) = 2$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_1) = ?$$



“Lower Bound”

“optimistic assignment”

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$y_2 = UB(y_2) = 2$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_1) = ?$$

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$y_2 = UB(y_2) = 2$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_1) = ?$$

$$y_1 + 2 \cdot 2 + 3 \cdot 4 \geq 17$$

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$y_2 = UB(y_2) = 2$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_1) = ?$$

$$y_1 + 2 \cdot 2 + 3 \cdot 4 \geq 17$$

$$y_1 + 16 \geq 17$$


$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$y_2 = UB(y_2) = 2$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_1) = 1$$

$$y_1 + 2 \cdot 2 + 3 \cdot 4 \geq 17$$

$$y_1 + 16 \geq 17$$


$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{\cancel{0}, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

Value '0' removed
by propagation

$$y_2 = UB(y_2) = 2$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_1) = 1$$

$$y_1 + 2 \cdot 2 + 3 \cdot 4 \geq 17$$

$$y_1 + 16 \geq 17$$

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{\cancel{0}, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

Explanation
of the propagation



$$y_2 = UB(y_2) = 2$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_1) = 1$$

$$\langle y_2 \leq 2 \rangle \wedge \langle y_3 \leq 4 \rangle \Rightarrow \langle y_1 \geq 1 \rangle$$

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{1, 4\}$$

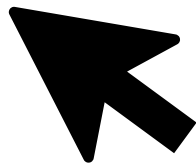
$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$y_1 = UB(y_2) = 4$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_2) = ?$$



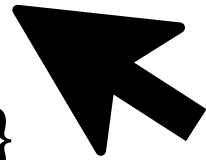
$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{1, 4\}$$

$$y_2 \in \{\cancel{0}, 2\}$$

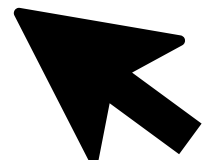
$$y_3 \in \{0, 1, 2, 3, 4\}$$

Value '0' removed
by propagation



$$y_1 = UB(y_2) = 4$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_2) = 2$$


$$4 + 2 \cdot y_2 + 3 \cdot 4 \geq 17$$

$$y_2 + 16 \geq 17$$

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{1, 4\}$$

$$y_2 \in \{\cancel{1}, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$



Explanation?

$$y_1 = UB(y_2) = 4$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_2) = 2$$

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{1, 4\}$$

$$y_2 \in \{\cancel{1}, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

Explanation



$$y_1 = UB(y_2) = 4$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_2) = 2$$

$$\langle y_1 \leq 4 \rangle \wedge \langle y_3 \leq 4 \rangle \Rightarrow \langle y_2 \geq 1 \rangle$$

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{1, 4\}$$

$$y_2 \in \{2\}$$

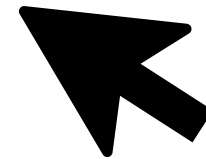
$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{1, 4\}$$

$$y_2 \in \{2\}$$

$$y_3 \in \{\cancel{0}, \cancel{1}, \cancel{2}, 3, 4\}$$



Values '0', '1', '2' removed
by propagation with explanation

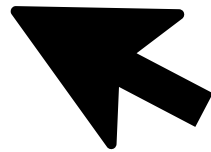
$$\langle y_1 \leq 4 \rangle \wedge \langle y_2 \leq 2 \rangle \Rightarrow \langle y_3 \geq 3 \rangle$$

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{\cancel{8}, 1, 4\}$$

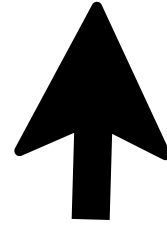
$$y_2 \in \{\cancel{8}, 2\}$$

$$y_3 \in \{\cancel{8}, \cancel{1}, \cancel{2}, 3, 4\}$$



Propagation shrinks domains

$$f(\{0, 1, 4\}, \{0, 2\}, \{0, 1, 2, 3, 4\}) = (\{1, 4\}, \{2\}, \{3, 4\})$$



$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{\cancel{0}, 1, 4\}$$

$$y_2 \in \{\cancel{0}, 2\}$$

$$y_3 \in \{\cancel{0}, \cancel{1}, \cancel{2}, 3, 4\}$$

Propagator as a function

$$f(\{0, 1, 4\}, \{0, 2\}, \{0, 1, 2, 3, 4\}) = (\{1, 4\}, \{2\}, \{3, 4\})$$

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{\cancel{0}, 1, 4\}$$

$$y_2 \in \{\cancel{0}, 2\}$$

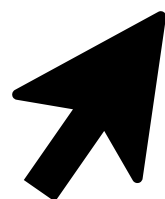
$$y_3 \in \{\cancel{0}, \cancel{1}, \cancel{2}, 3, 4\}$$

$$\langle y_2 \leq 2 \rangle \wedge \langle y_3 \leq 4 \rangle \Rightarrow \langle y_1 \geq 1 \rangle$$

$$\langle y_1 \leq 4 \rangle \wedge \langle y_3 \leq 4 \rangle \Rightarrow \langle y_2 \geq 1 \rangle$$

$$\langle y_1 \leq 4 \rangle \wedge \langle y_2 \leq 2 \rangle \Rightarrow \langle y_3 \geq 3 \rangle$$

**Propagation explained
with three steps**



$$\sum w_i \cdot y_i \geq k$$

(assume $w_i \geq 0$)

Propagator

$$y_i \geq \left\lceil \frac{k - \sum_{j \neq i} w_j \cdot UB(y_j)}{w_i} \right\rceil$$

$$\sum w_i \cdot y_i \geq k$$

(assume $w_i \geq 0$)

set other variables
to the optimistic assignment

Propagator

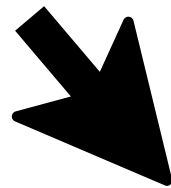


$$y_i \geq \left\lceil \frac{k - \sum_{j \neq i} w_j \cdot \boxed{UB(y_j)}}{w_i} \right\rceil$$

units that
must be covered by
the term $w_i \cdot y_i$

$$\sum w_i \cdot y_i \geq k$$

(assume $w_i \geq 0$)

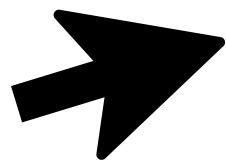


Propagator

$$y_i \geq \frac{\left[k - \sum_{j \neq i} w_j \cdot UB(y_j) \right]}{w_i}$$

Think about
the general case

$$w_i \in \mathbb{Z}$$



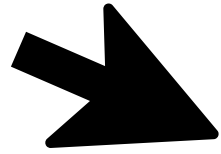
$$\sum w_i \cdot y_i \geq k$$

(assume $w_i \geq 0$)

Propagator

$$y_i \geq \left\lceil \frac{k - \sum_{j \neq i} w_j \cdot UB(y_j)}{w_i} \right\rceil$$

Defines feasibility



Constraint



Propagator



Propagation
Algorithm

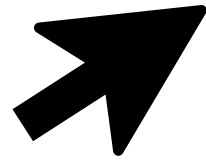
Constraint



Propagator



Propagation
Algorithm



Defines rules
to shrink domains
without changing feasibility

Constraint

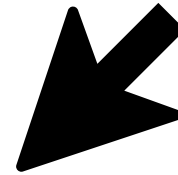


Propagator



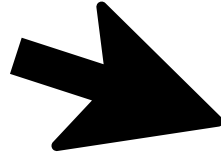
Propagation
Algorithm

Defines **how**
the inference is computed



Data structures
extremely
important!

$$\sum w_i \cdot y_i \geq k$$



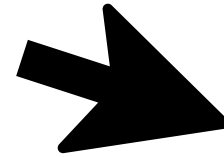
In practice,
propagation algorithms can be complex...

Propagator

$$y_i \geq \left\lceil \frac{k - \sum_{j \neq i} w_j \cdot UB(y_j)}{w_i} \right\rceil$$

$$\sum w_i \cdot y_i \geq k$$

In practice,
propagation algorithms can be complex...



Handle $w_i \in \mathbb{Z}$

Propagator

$$y_i \geq \left\lceil \frac{k - \sum_{j \neq i} w_j \cdot UB(y_j)}{w_i} \right\rceil$$

$$\sum w_i \cdot y_i \geq k$$

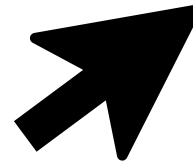
In practice,
propagation algorithms can be complex...

Propagator

$$y_i \geq \left\lceil \frac{k - \sum_{j \neq i} w_j \cdot UB(y_j)}{w_i} \right\rceil$$

Handle $w_i \in \mathbb{Z}$

Incremental propagation



$$\sum w_i \cdot y_i \geq k$$

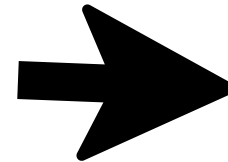
In practice,
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Propagator

$$y_i \geq \left\lceil \frac{k - \sum_{j \neq i} w_j \cdot UB(y_j)}{w_i} \right\rceil$$

Handle $w_i \in \mathbb{Z}$

Incremental propagation



Rounding errors

$$\sum w_i \cdot y_i \geq k$$

In practice,
propagation algorithms can be complex...

Propagator

$$y_i \geq \left\lceil \frac{k - \sum_{j \neq i} w_j \cdot UB(y_j)}{w_i} \right\rceil$$

Handle $w_i \in \mathbb{Z}$

Incremental propagation

Rounding errors

Special cases, e.g., $w_i = 1$



$$\sum w_i \cdot y_i \geq k$$

In practice,
propagation algorithms can be complex...

Propagator

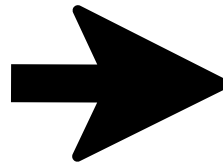
$$y_i \geq \left\lceil \frac{k - \sum_{j \neq i} w_j \cdot UB(y_j)}{w_i} \right\rceil$$

Handle $w_i \in \mathbb{Z}$

Incremental propagation

Rounding errors

Special cases, e.g., $w_i = 1$



Many things can go wrong

$$f(\{0, 1, 4\}, \{0, 2\}, \{0, 1, 2, 3, 4\}) = (\{1, 4\}, \{2\}, \{3, 4\})$$

$$y_1 + 2y_2 + 3y_3 \geq 17$$



How to trust this propagation?

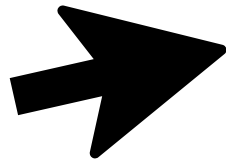
$$f(\{0, 1, 4\}, \{0, 2\}, \{0, 1, 2, 3, 4\}) = (\{1, 4\}, \{2\}, \{3, 4\})$$

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$\langle y_2 \leq 2 \rangle \wedge \langle y_3 \leq 4 \rangle \Rightarrow \langle y_1 \geq 1 \rangle$$

$$\langle y_1 \leq 4 \rangle \wedge \langle y_3 \leq 4 \rangle \Rightarrow \langle y_2 \geq 1 \rangle$$

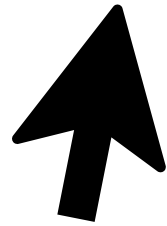
$$\langle y_1 \leq 4 \rangle \wedge \langle y_2 \leq 2 \rangle \Rightarrow \langle y_3 \geq 3 \rangle$$



Much simpler task:
Check the explanations!

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$\langle y_1 \leq 4 \rangle \wedge \langle y_3 \leq 4 \rangle \Rightarrow \langle y_2 \geq 1 \rangle$$



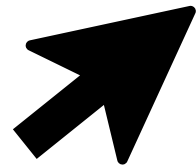
Is the explanation sound?

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$\langle y_1 \leq 4 \rangle \wedge \langle y_3 \leq 4 \rangle \Rightarrow \langle y_2 \geq 1 \rangle$$



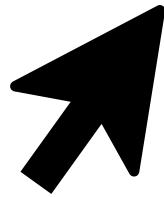
$$\langle y_1 \leq 4 \rangle \wedge \langle y_2 \leq 0 \rangle \wedge \langle y_3 \leq 4 \rangle \Rightarrow \perp$$



**Reduce verification
to feasibility checking**

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$\langle y_1 \leq 4 \rangle \wedge \langle y_2 \leq 0 \rangle \wedge \langle y_3 \leq 4 \rangle \Rightarrow \perp$$



Much simpler task!

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$\langle y_1 \leq 4 \rangle \wedge \langle y_2 \leq 0 \rangle \wedge \langle y_3 \leq 4 \rangle \Rightarrow \perp$$

$$4 + 2 \cdot 0 + 3 \cdot 4 \geq 17$$



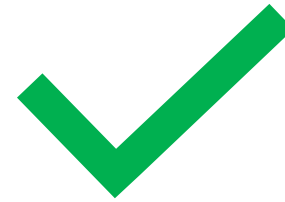
**Assume optimistic assignment
based on explanation**

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$\langle y_1 \leq 4 \rangle \wedge \langle y_2 \leq 0 \rangle \wedge \langle y_3 \leq 4 \rangle \Rightarrow \perp$$

$$4 + 2 \cdot 0 + 3 \cdot 4 \geq 17$$

$$16 \geq 17$$



Derive conflict to verify explanation!

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$\langle y_1 \leq 4 \rangle \wedge \langle y_2 \leq 0 \rangle \wedge \langle y_3 \leq 4 \rangle \Rightarrow \perp$$

$$4 + 2 \cdot 0 + 3 \cdot 4 \geq 17$$

$$16 \geq 17$$



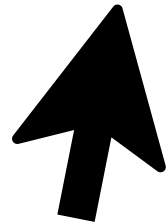
Checker: verifies an explanation

- 1. Independent of the propagator**
- 2. Much simpler logic → formally verified**

Another Example

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$\langle y_1 \leq 8 \rangle \wedge \langle y_3 \leq 3 \rangle \Rightarrow \langle y_2 \geq 1 \rangle$$



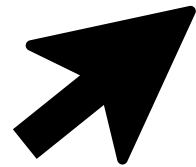
Is the explanation sound?

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$\langle y_1 \leq 8 \rangle \wedge \langle y_3 \leq 3 \rangle \Rightarrow \langle y_2 \geq 1 \rangle$$



$$\langle y_1 \leq 8 \rangle \wedge \langle y_2 \leq 0 \rangle \wedge \langle y_3 \leq 3 \rangle \Rightarrow \perp$$

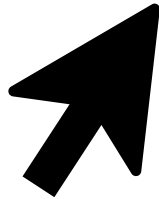


**Reduce verification
to feasibility checking**

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$\langle y_1 \leq 8 \rangle \wedge \langle y_2 \leq 0 \rangle \wedge \langle y_3 \leq 3 \rangle \Rightarrow \perp$$

$$8 + 2 \cdot 0 + 3 \cdot 3 \geq 17$$



$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$\langle y_1 \leq 8 \rangle \wedge \langle y_2 \leq 0 \rangle \wedge \langle y_3 \leq 3 \rangle \Rightarrow \perp$$

$$8 + 2 \cdot 0 + 3 \cdot 3 \geq 17$$

$$17 \geq 17$$



No conflict, explanation not justified!

Constraint



Propagator



Propagation
Algorithm

Constraint

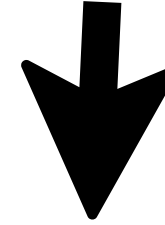


Propagator



Propagation
Algorithm

Justifies
propagation



Explanations

Constraint



Propagator



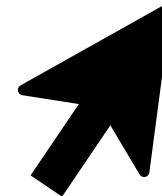
Explanations



Propagation
Algorithm



Checker



Verifies explanations

Propagation for the Maximum Constraint

$$x = \max(y, z)$$

Propagation for the Maximum Constraint

$$x = \max(y, z)$$

x, y, z are integer interval variables

Propagation for the Maximum Constraint

$$x = \max(y, z)$$

x, y, z are integer interval variables

Feasible solutions

$$x = 5$$

$$y = 0$$

$$z = 5$$

$$x = 10$$

$$y = 10$$

$$z = 10$$

$$x = 0$$

$$y = -5$$

$$z = 0$$

Propagation for the Maximum Constraint

$$x = \max(y, z)$$

x, y, z are integer interval variables

Feasible solutions

$$x = 5$$

$$y = 0$$

$$z = 5$$

$$x = 10$$

$$y = 10$$

$$z = 10$$

$$x = 0$$

$$y = -5$$

$$z = 0$$

Infeasible solutions

$$x = 5$$

$$y = 10$$

$$z = 3$$

$$x = 10$$

$$y = 7$$

$$z = 9$$

$$x = 0$$

$$y = 1$$

$$z = 2$$

Propagation for the Maximum Constraint

$$x = \max(y, z)$$

x, y, z are integer interval variables

Feasible solutions

$$\begin{aligned}x &= 5 \\y &= 0 \\z &= 5\end{aligned}$$

$$\begin{aligned}x &= 10 \\y &= 10 \\z &= 10\end{aligned}$$

$$\begin{aligned}x &= 0 \\y &= -5 \\z &= 0\end{aligned}$$

Devise the
propagation
rules!

Infeasible solutions

$$\begin{aligned}x &= 5 \\y &= 10 \\z &= 3\end{aligned}$$

$$\begin{aligned}x &= 10 \\y &= 7 \\z &= 9\end{aligned}$$

$$\begin{aligned}x &= 0 \\y &= 1 \\z &= 2\end{aligned}$$

Example

$$x = \max(y, z)$$

Before propagation

$$x \in [0, 20]$$



$$y \in [0, 30]$$



$$z \in [0, 40]$$



After propagation

?



Example

$$x = \max(y, z)$$

Before propagation

After propagation

$x \in [0, 20]$

0  20

0  20

$y \in [0, 30]$

0  30

0  20

$z \in [0, 40]$

0  40

0  20

Example

$$x = \max(y, z)$$

Before propagation

$$x \in [0, 20]$$



$$y \in [0, 30]$$



$$z \in [0, 40]$$



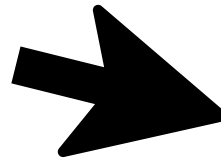
After propagation



Explanations

$$\langle x \leq 20 \rangle \Rightarrow \langle y \leq 20 \rangle$$

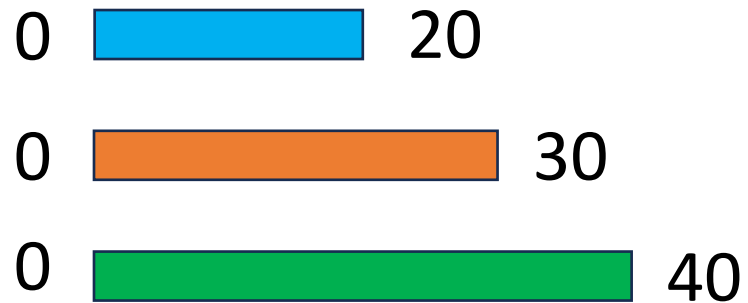
$$\langle x \leq 20 \rangle \Rightarrow \langle z \leq 20 \rangle$$



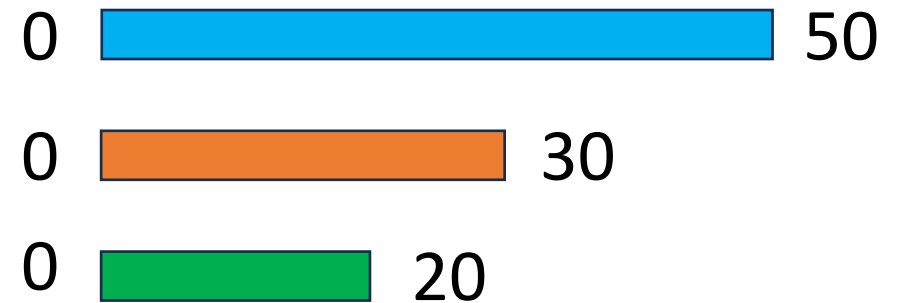
Devise the propagation rules!

$$x = \max(y, z)$$

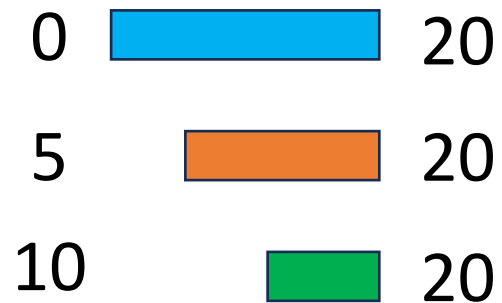
Scenario 1



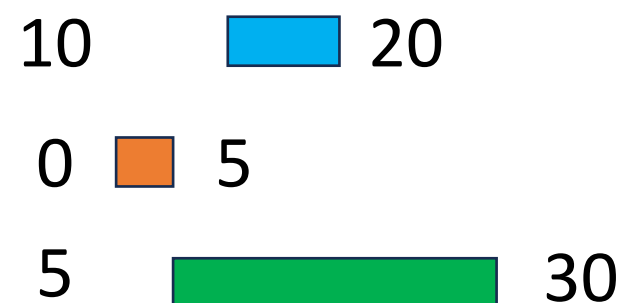
Scenario 3



Scenario 2

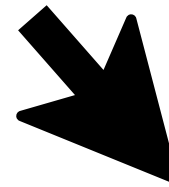
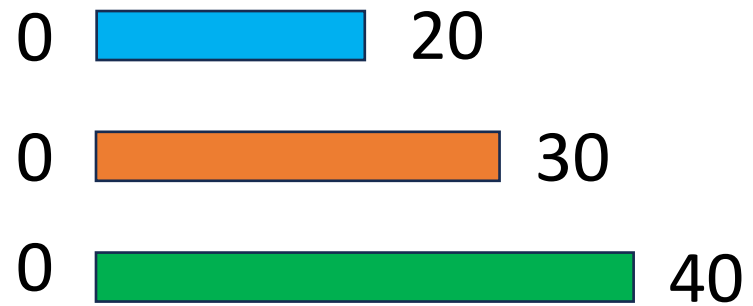


Scenario 4



$$x = \max(y, z)$$

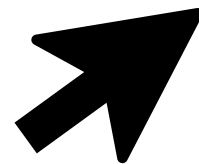
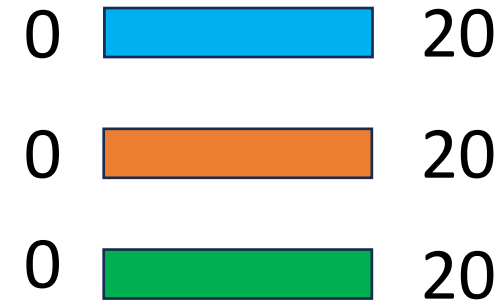
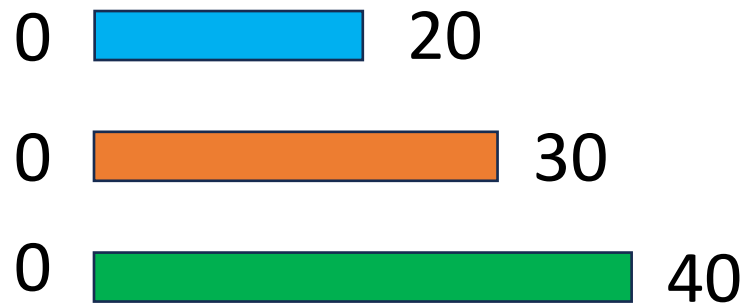
Scenario 1



Rule?

$$x = \max(y, z)$$

Scenario 1



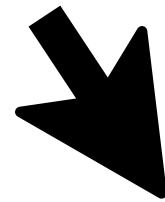
$$\langle x \leq k \rangle \Rightarrow \langle y \leq k \rangle$$

$$\langle x \leq k \rangle \Rightarrow \langle z \leq k \rangle$$

$$x = \max(y, z)$$

Scenario 2

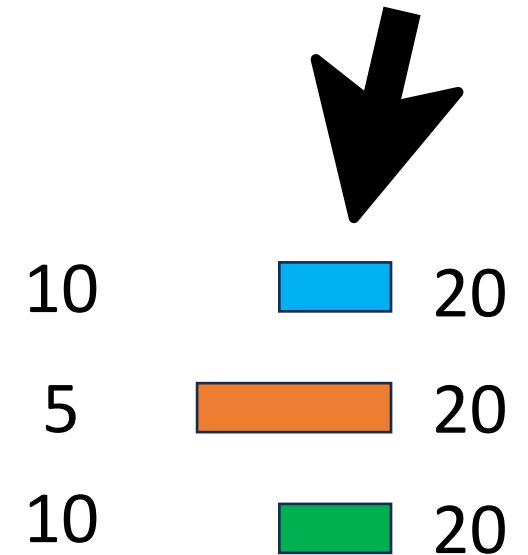
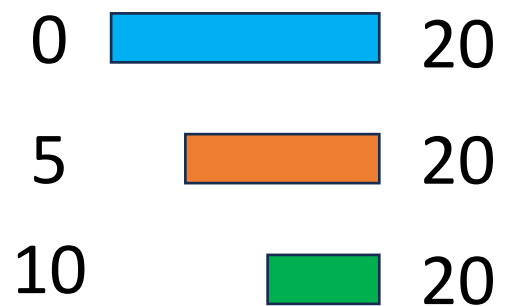
0		20
5		20
10		20



Propagation?

$$x = \max(y, z)$$

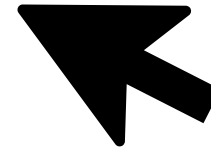
Scenario 2



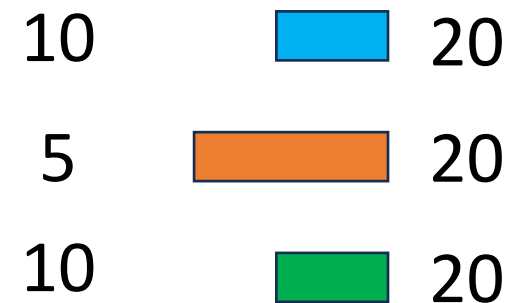
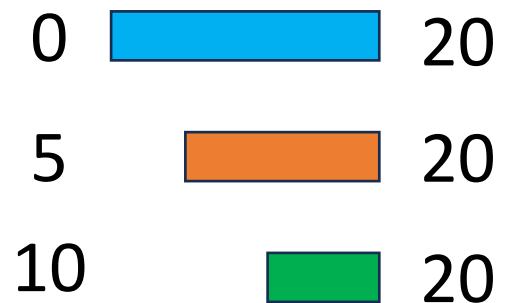
$$x = \max(y, z)$$

$$\langle y \geq k \rangle \Rightarrow \langle x \geq k \rangle$$

$$\langle z \geq k \rangle \Rightarrow \langle x \geq k \rangle$$

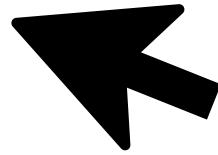


Scenario 2

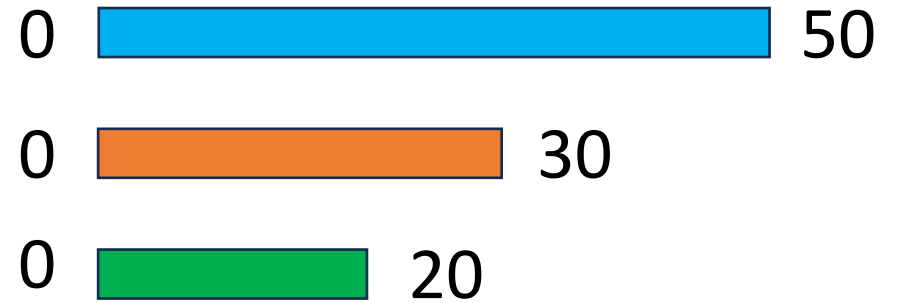


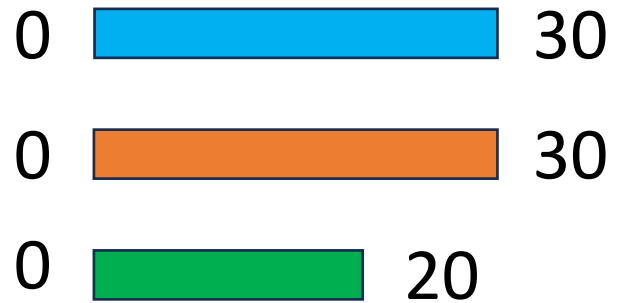
$$x = \max(y, z)$$

Propagation?



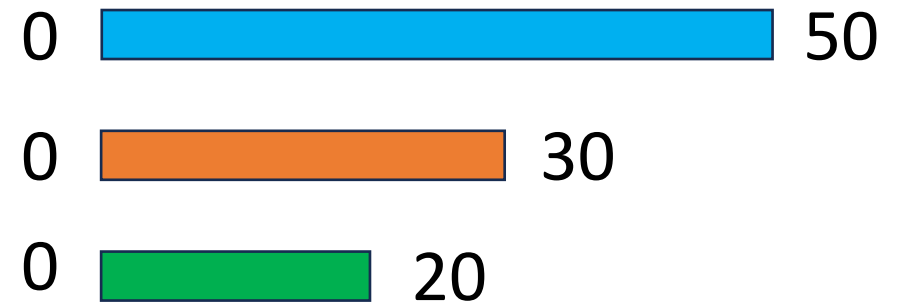
Scenario 3





$$x = \max(y, z)$$

Scenario 3



$$x = \max(y, z)$$

Scenario 3

0  30

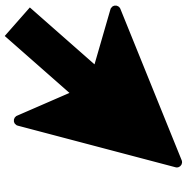
0  30

0  20

0  50

0  30

0  20



$$\langle y \leq a \rangle \wedge \langle z \leq b \rangle \Rightarrow \langle x \leq \max(a, b) \rangle$$

$$x = \max(y, z)$$

0  30

0  30

0  20

Scenario 3

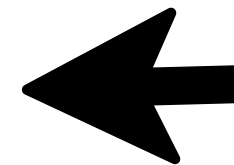
0  50

0  30

0  20

$$\langle y \leq a \rangle \wedge \langle z \leq b \rangle \Rightarrow \langle x \leq \max(a, b) \rangle$$

$$\langle y \leq k \rangle \wedge \langle z \leq k \rangle \Rightarrow \langle x \leq k \rangle$$



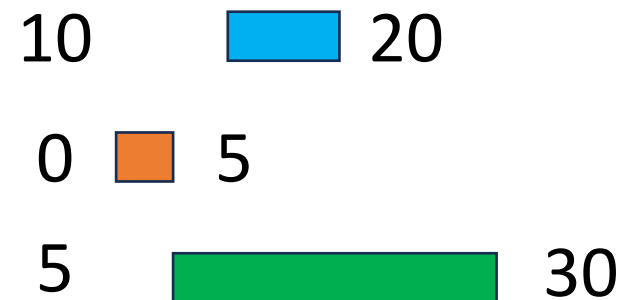
Lifted
(more general)
Explanation

$$x = \max(y, z)$$

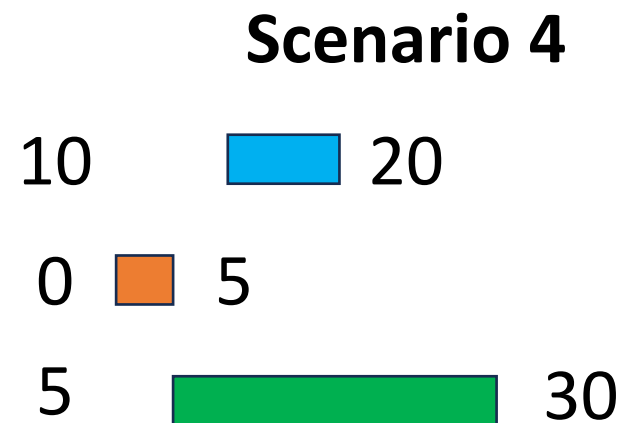
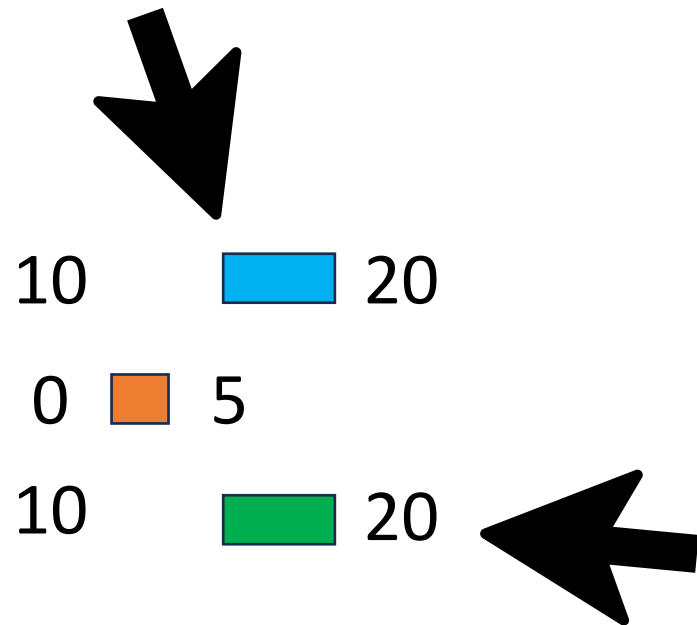


Propagation?

Scenario 4

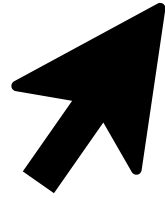


$$x = \max(y, z)$$



$$x = \max(y, z)$$

$$\langle x \geq a + 1 \rangle \wedge \langle y \leq a \rangle \Rightarrow \langle x = z \rangle$$



10  20

0  5

10  20

Scenario 4

10  20

0  5

5  30

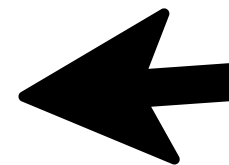
$$x = \max(y, z)$$

$$\langle x \leq k \rangle \Rightarrow \langle y \leq k \rangle$$

$$\langle x \leq k \rangle \Rightarrow \langle z \leq k \rangle$$

$$\langle y \leq k \rangle \wedge \langle z \leq k \rangle \Rightarrow \langle x \leq k \rangle$$

Checker?



$$\langle y \geq k \rangle \Rightarrow \langle x \geq k \rangle$$

$$\langle z \geq k \rangle \Rightarrow \langle x \geq k \rangle$$

$$\langle x \geq a + 1 \rangle \wedge \langle y \leq a \rangle \Rightarrow \langle x = z \rangle$$

$$x = \max(y, z)$$

$$\langle x \leq k \rangle \Rightarrow \langle y \leq k \rangle$$

$$\langle x \leq k \rangle \Rightarrow \langle z \leq k \rangle$$

$$\langle y \leq k \rangle \wedge \langle z \leq k \rangle \Rightarrow \langle x \leq k \rangle$$

Checker



$$\langle y \geq k \rangle \Rightarrow \langle x \geq k \rangle$$

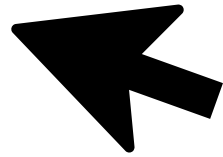
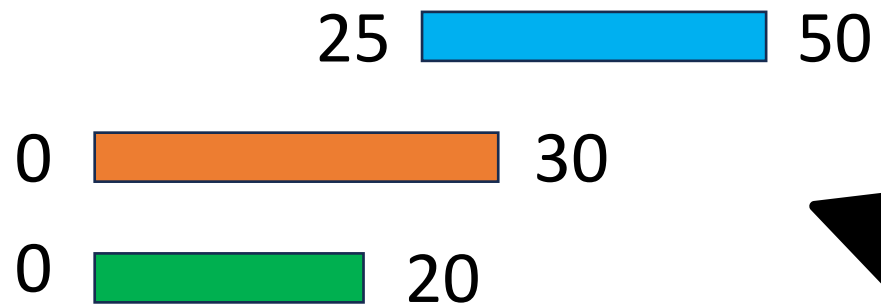
$$\langle z \geq k \rangle \Rightarrow \langle x \geq k \rangle$$

**Reduce to
conflict
detection!**

$$\langle x \geq a + 1 \rangle \wedge \langle y \leq a \rangle \Rightarrow \langle x = z \rangle$$

$$x = \max(y, z)$$

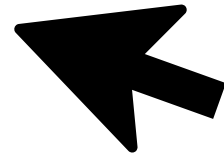
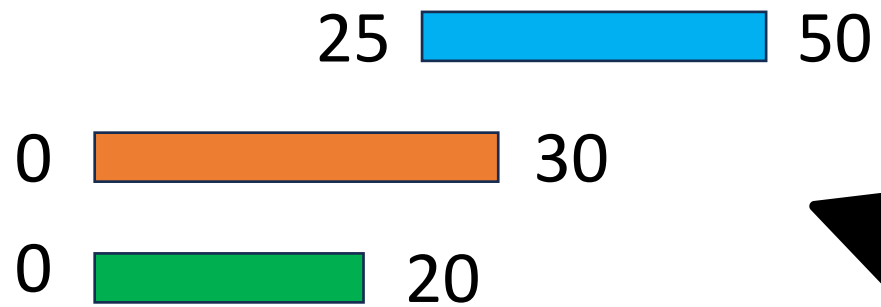
Scenario 1



Conflict?

$$x = \max(y, z)$$

Scenario 1

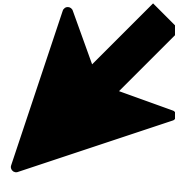


No conflict!

$$x = \max(y, z)$$

Conflict?

Scenario 1



35  50

0  30

0  20

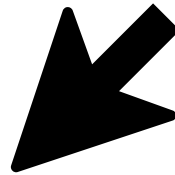
$$x = \max(y, z)$$

Scenario 1

35  50

0  30

0  20

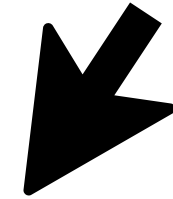


Conflict?

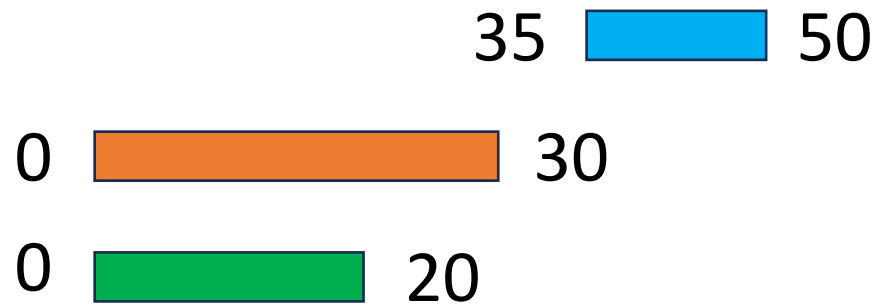
Yes!

$$x = \max(y, z)$$

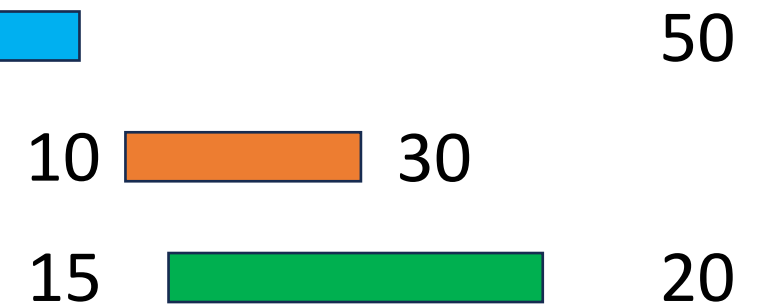
Conflicts



Scenario 1



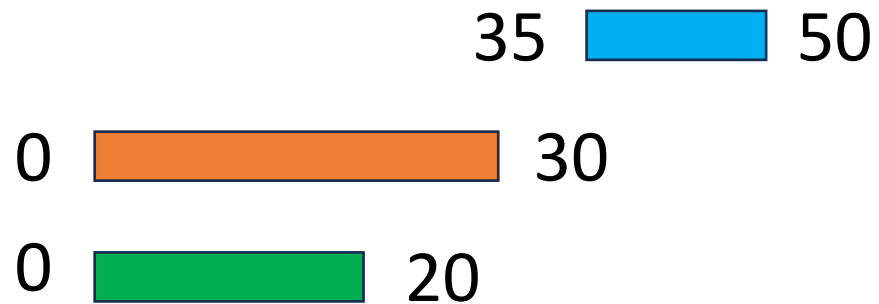
Scenario 2



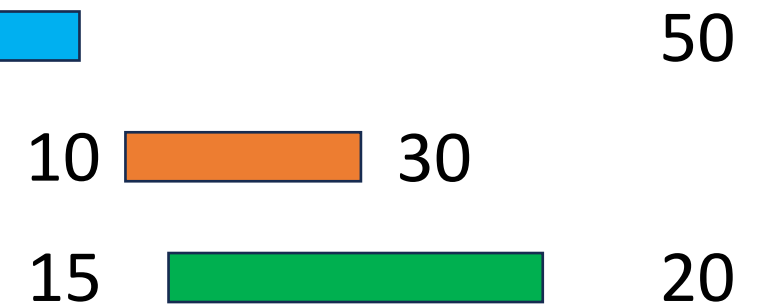
$$x = \max(y, z)$$

Conflicts

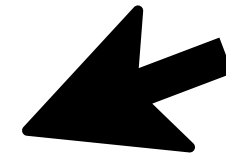
Scenario 1



Scenario 2



Checker rule?



$$x = \max(y, z)$$

Checker

$$*a = \max(LB(y), LB(z))*$$

$$*b = \max(UB(y), UB(z))*$$

$$[a, b] \cap [LB(x), UB(x)] = \emptyset$$

$$x = \max(y, z)$$

$$\langle x \leq k \rangle \Rightarrow \langle y \leq k \rangle$$

$$\langle y \leq k \rangle \wedge \langle z \leq k \rangle \Rightarrow \langle x \leq k \rangle$$

$$\langle x \leq k \rangle \Rightarrow \langle z \leq k \rangle$$

Checker $\rightarrow$ one rule!

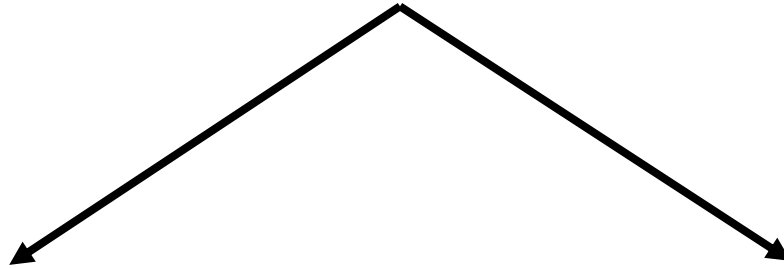
$$\langle y \geq k \rangle \Rightarrow \langle x \geq k \rangle$$

$$\langle x \geq a + 1 \rangle \wedge \langle y \leq a \rangle \Rightarrow \langle x = z \rangle$$

$$\langle z \geq k \rangle \Rightarrow \langle x \geq k \rangle$$

Up next...

All-Different



Decomposition

**Main idea for
checker and propagation**

AllDifferent Constraint

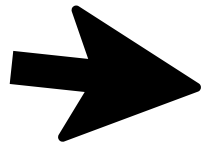
“all variables must take distinct values”

$\text{All-Different}(x_1, x_2, x_3, x_4)$

AllDifferent Constraint

“all variables must take distinct values”

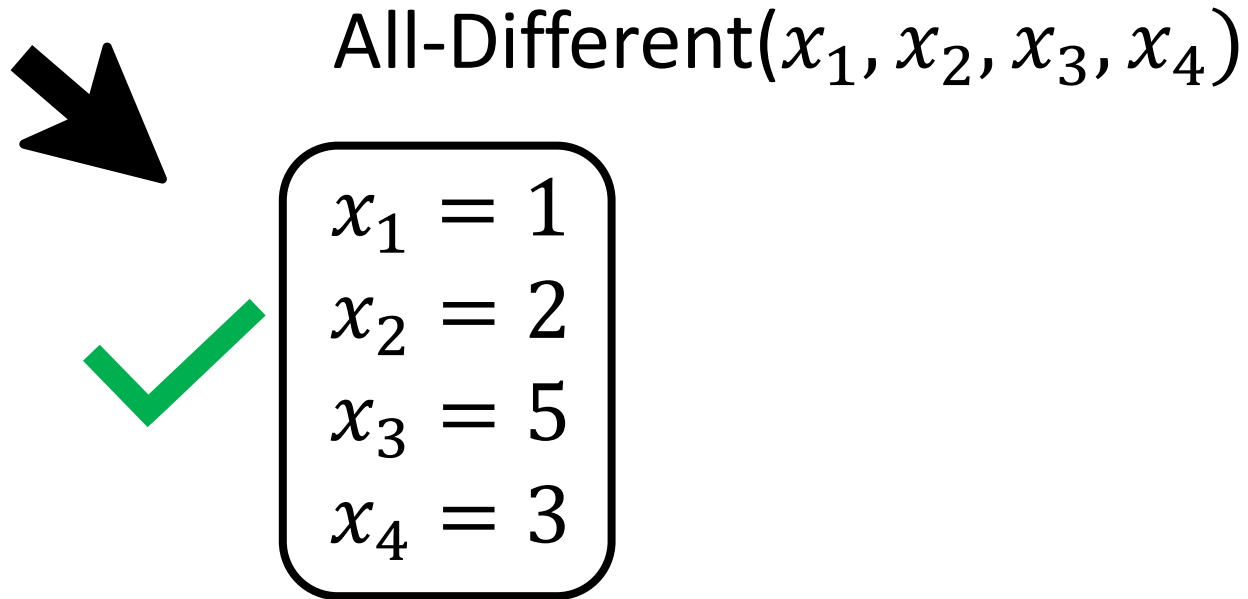
$\text{All-Different}(x_1, x_2, x_3, x_4)$



Common in assignment problems!

AllDifferent Constraint

“all variables must take distinct values”



AllDifferent Constraint

“all variables must take distinct values”

All-Different(x_1, x_2, x_3, x_4)



$x_1 = 1$
 $x_2 = 2$
 $x_3 = 5$
 $x_4 = 3$

$x_1 = 1$
 $x_2 = 1$
 $x_3 = 5$
 $x_4 = 3$



AllDifferent Constraint

$$x_i \in D_i \subset \mathbb{N}$$

All-Different($x_1, x_2, \dots, x_n$)

How to propagate?

$$x_1, x_2, x_3, x_4 \in \{1, 2, 3, 4\}$$

“all variables must take distinct values”

$$x_i \neq x_j$$

$$x_1 \neq x_2$$

$$x_1 \neq x_3 \quad x_2 \neq x_3$$

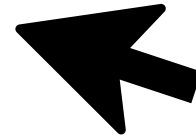
$$x_1 \neq x_4 \quad x_2 \neq x_4 \quad x_3 \neq x_4$$

$$x_1 \in \{1, 2, 3, 4\}$$

$$x_2 \in \{1, 2, 3, 4\}$$

$$x_3 \in \{1, 2, 3, 4\}$$

$$x_4 \in \{1, 2, 3, 4\}$$



No propagation

$$x_1 \neq x_2$$

$$x_1 \neq x_3 \quad x_2 \neq x_3$$

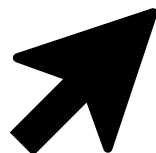
$$x_1 \neq x_4 \quad x_2 \neq x_4 \quad x_3 \neq x_4$$

$$x_1 \in \{1, 2, 3, 4\}$$

$$x_2 \in \{1, 2, 3, 4\}$$

$$x_3 \in \{1, 2, 3, 4\}$$

$$x_4 \in \{1\}$$



Propagate?

$$x_1 \neq x_2$$

$$x_1 \neq x_3 \quad x_2 \neq x_3$$

$$x_1 \neq x_4 \quad x_2 \neq x_4 \quad x_3 \neq x_4$$

$$\begin{array}{l}
 x_1 \in \{\cancel{1}, 2, 3, 4\} \\
 x_2 \in \{1, 2, 3, 4\} \\
 x_3 \in \{1, 2, 3, 4\} \\
 x_4 \in \{1\}
 \end{array}
 \quad \blacktriangleleft \quad \langle x_4 = 1 \rangle \Rightarrow \langle x_1 \neq 1 \rangle$$

$$x_1 \neq x_2$$

$$x_1 \neq x_3 \quad x_2 \neq x_3$$

$$x_1 \neq x_4 \quad x_2 \neq x_4 \quad x_3 \neq x_4$$

$$x_1 \in \{\cancel{1}, 2, 3, 4\}$$

$$x_2 \in \{\cancel{1}, 2, 3, 4\}$$

$$x_3 \in \{\cancel{1}, 2, 3, 4\}$$

$$x_4 \in \{1\}$$

$$\langle x_i = k \rangle \Rightarrow \langle x_j \neq k \rangle$$



$$x_1 \neq x_2$$

$$x_1 \neq x_3 \quad x_2 \neq x_3$$

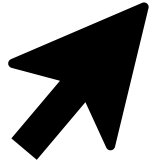
$$x_1 \neq x_4 \quad x_2 \neq x_4 \quad x_3 \neq x_4$$

$$x_1 \in \{1, 3\}$$

$$x_2 \in \{1, 3\}$$

$$x_3 \in \{1, 2, 4\}$$

$$x_4 \in \{1, 2\}$$



Propagate?

$$x_1 \neq x_2$$

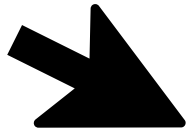
$$x_1 \neq x_3 \quad x_2 \neq x_3$$

$$x_1 \neq x_4 \quad x_2 \neq x_4 \quad x_3 \neq x_4$$

$$\begin{aligned}x_1 &\in \{1, 3\} \\x_2 &\in \{1, 3\} \\x_3 &\in \{1, 2, 4\} \\x_4 &\in \{1, 2\}\end{aligned}$$

$$\langle x_i = k \rangle \Rightarrow \langle x_j \neq k \rangle$$

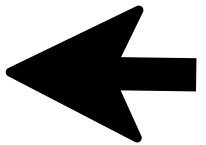
No propagation!



$$x_1 \neq x_2$$

$$x_1 \neq x_3 \quad x_2 \neq x_3$$

$$x_1 \neq x_4 \quad x_2 \neq x_4 \quad x_3 \neq x_4$$

$$\begin{aligned}x_1 &\in \{1, 3\} \\x_2 &\in \{1, 3\} \\x_3 &\in \{1, 2, 4\} \\x_4 &\in \{1, 2\}\end{aligned}$$


'4' must be assigned here

$$x_1 \neq x_2$$

$$x_1 \neq x_3 \quad x_2 \neq x_3$$

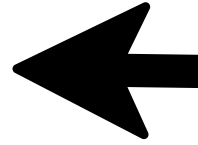
$$x_1 \neq x_4 \quad x_2 \neq x_4 \quad x_3 \neq x_4$$

$$x_1 \in \{1, 3\}$$

$$x_2 \in \{1, 3\}$$

$$x_3 \in \{4\}$$

$$x_4 \in \{1, 2\}$$



'2' must be assigned here

$$x_1 \neq x_2$$

$$x_1 \neq x_3 \quad x_2 \neq x_3$$

$$x_1 \neq x_4 \quad x_2 \neq x_4 \quad x_3 \neq x_4$$

$$x_1 \in \{1, 3\}$$

$$x_2 \in \{1, 3\}$$

$$x_3 \in \{4\}$$

$$x_4 \in \{2\}$$

$$x_1 \neq x_2$$

$$x_1 \neq x_3 \quad x_2 \neq x_3$$

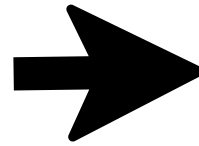
$$x_1 \neq x_4 \quad x_2 \neq x_4 \quad x_3 \neq x_4$$

$$x_1 \in \{1, 3\}$$

$$x_2 \in \{1, 3\}$$

$$x_3 \in \{4\}$$

$$x_4 \in \{2\}$$



**Stronger reasoning
by combining constraints!**

$$x_1 \neq x_2$$

$$x_1 \neq x_3 \quad x_2 \neq x_3$$

$$x_1 \neq x_4 \quad x_2 \neq x_4 \quad x_3 \neq x_4$$

$$x_1 \in \{1, 2, 3\}$$

$$x_2 \in \{1, 2, 3\}$$

$$x_3 \in \{1, 2, 3\}$$

$$x_4 \in \{1, 2\}$$



$$x_1 \neq x_2$$

$$x_1 \neq x_3 \quad x_2 \neq x_3$$

$$x_1 \neq x_4 \quad x_2 \neq x_4 \quad x_3 \neq x_4$$

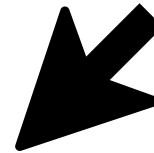
$$x_1 \in \{1, 2, 3\}$$

$$x_2 \in \{1, 2, 3\}$$

$$x_3 \in \{1, 2, 3\}$$

$$x_4 \in \{1, 2\}$$

no propagation
even though
infeasible!



$$x_1 \neq x_2$$

$$x_1 \neq x_3 \quad x_2 \neq x_3$$

$$x_1 \neq x_4 \quad x_2 \neq x_4 \quad x_3 \neq x_4$$

Variant 1:
Decomposition
for All-Different

All-Different, Version 1: Decomposition

$$x_1, x_2, x_3, x_4 \in \{1, 2, 3, 4\}$$

“all variables must take distinct values”

$$x_i \neq x_j$$

Pros?

$$x_1 \neq x_2$$

$$x_1 \neq x_3$$

$$x_2 \neq x_3$$

$$x_1 \neq x_4$$

$$x_2 \neq x_4$$

$$x_3 \neq x_4$$

Cons?

All-Different, Version 1: Decomposition

$$x_1, x_2, x_3, x_4 \in \{1, 2, 3, 4\}$$

“all variables must take distinct values”

$$x_i \neq x_j$$

Computationally
inexpensive

$$x_1 \neq x_2$$

$$x_1 \neq x_3$$

$$x_2 \neq x_3$$

$$x_1 \neq x_4$$

$$x_2 \neq x_4$$

$$x_3 \neq x_4$$

Misses
propagations

Variant 2:
Domain Consistency
for All-Different

All-Different: Main idea for stronger propagation

$$x_1 \in \{1, 3\}$$

$$x_2 \in \{1, 3\}$$

$$x_3 \in \{1, 2, 3, 4, 5, 6\}$$

$$x_4 \in \{1, 2\}$$

All-Different: Main idea for stronger propagation

$$x_1 \in \{1, 3\}$$

$$x_2 \in \{1, 3\}$$

$$x_3 \in \{1, 2, 3, 4, 5, 6\}$$

$$x_4 \in \{1, 2\}$$

All-Different: Main idea for stronger propagation

$$x_1 \in \{1, 3\}$$

$$x_2 \in \{1, 3\}$$

$$x_3 \in \{1, 2, 3, 4, 5, 6\}$$

$$x_4 \in \{1, 2\}$$

$$H = \{x_1, x_2, x_3\}$$

$$\begin{aligned} \text{Union of domains in } H \\ = \{1, 2, 3\} \end{aligned}$$

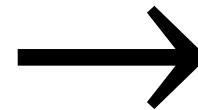
What can we conclude about x_3 ?

All-Different: Main idea for stronger propagation

$$\begin{aligned}x_1 &\in \{1, 3\} \\x_2 &\in \{1, 3\} \\x_3 &\in \{\cancel{1}, \cancel{2}, \cancel{3}, 4, 5, 6\} \\x_4 &\in \{1, 2\}\end{aligned}$$

$$H = \{x_1, x_2, x_3\}$$

$$\begin{aligned}\text{Union of domains in } H \\&= \{1, 2, 3\}\end{aligned}$$



$$x_3 \neq \{1, 2, 3\}$$

(variables in H are occupying values $\{1, 2, 3\}$,
so x_3 cannot take any of those values)

All-Different: Main idea for stronger propagation

$$\begin{aligned} x_1 &\in \{1, 3\} \\ x_2 &\in \{1, 3\} \\ x_3 &\in \{\cancel{1}, \cancel{2}, \cancel{3}, 4, 5, 6\} \\ x_4 &\in \{1, 2\} \end{aligned} \quad \blacktriangleleft$$

$$H = \{x_1, x_2\}$$

$$\begin{aligned} \text{Union of domains in } H \\ &= \{1, 3\} \end{aligned}$$

What can we conclude about x_4 ?

All-Different: Main idea for stronger propagation

$$\begin{aligned}x_1 &\in \{1, 3\} \\x_2 &\in \{1, 3\} \\x_3 &\in \{\cancel{1}, \cancel{2}, \cancel{3}, 4, 5, 6\} \\x_4 &\in \{\cancel{1}, 2\}\end{aligned}$$

$$H = \{x_1, x_2\}$$

$$\begin{aligned}\text{Union of domains in } H \\&= \{1, 3\}\end{aligned}$$



$$x_4 \neq 1$$

Another Example: Propagate?

$$x_1 \in \{1, 2, 3\}$$

$$x_2 \in \{1, 2, 3\}$$

$$x_3 \in \{1, 2, 3\}$$

$$x_4 \in \{1, 2, 3\}$$

Hall set

$$x_1 \in \{1, 2, 3\}$$

$$x_2 \in \{1, 2, 3\}$$

$$x_3 \in \{1, 2, 3\}$$

$$x_4 \in \{\cancel{1}, \cancel{2}, \cancel{3}\}$$


$$x_1 \in \{1, 2, 3\}$$

$$x_2 \in \{1, 2, 3\}$$

$$x_3 \in \{1, 2, 3\}$$

$$x_4 \in \{\cancel{1}, \cancel{2}, \cancel{3}\}$$

**Conflict
explanation?**



$$\langle x_1 \geq 1 \rangle \wedge \langle x_1 \leq 3 \rangle \wedge$$
$$\langle x_2 \geq 1 \rangle \wedge \langle x_2 \leq 3 \rangle \wedge$$
$$\langle x_3 \geq 1 \rangle \wedge \langle x_3 \leq 3 \rangle \wedge$$
$$\langle x_4 \geq 1 \rangle \wedge \langle x_4 \leq 3 \rangle$$

$\Rightarrow$

$\perp$

$$x_1 \in \{1, 2, 3\}$$

$$x_2 \in \{1, 2, 3\}$$

$$x_3 \in \{1, 2, 3\}$$

$$x_4 \in \{1, 2, 3\}$$

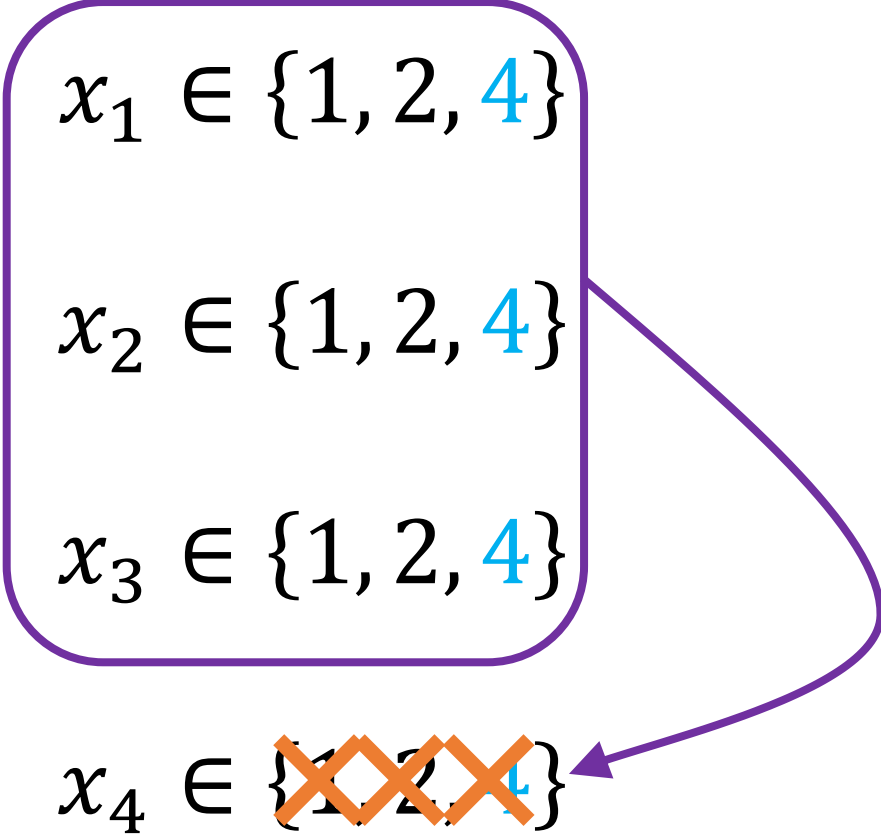
Example with Holes

Hall set

$$x_1 \in \{1, 2, 4\}$$

$$x_2 \in \{1, 2, 4\}$$

$$x_3 \in \{1, 2, 4\}$$

$$x_4 \in \{\cancel{1}, \cancel{2}, \cancel{4}\}$$


The diagram illustrates a Hall set for a matching problem. A purple rounded rectangle encloses the first three equations, with the label 'Hall set' to its left. A purple arrow points from the right side of this rectangle to the set notation of the fourth equation. In the fourth equation, the elements 1, 2, and 4 are each crossed out with a large orange 'X', while the number 4 remains in blue.

Example with Holes

$$\begin{aligned} &\langle x_1 \geq 1 \rangle \wedge \langle x_1 \neq 1 \rangle \wedge \langle x_1 \leq 4 \rangle \wedge \\ &\langle x_2 \geq 1 \rangle \wedge \langle x_2 \neq 1 \rangle \wedge \langle x_2 \leq 4 \rangle \wedge \\ &\langle x_3 \geq 1 \rangle \wedge \langle x_3 \neq 1 \rangle \wedge \langle x_3 \leq 4 \rangle \wedge \\ &\langle x_4 \geq 1 \rangle \wedge \langle x_4 \neq 1 \rangle \wedge \langle x_4 \leq 4 \rangle \\ &\Rightarrow \\ &\perp \end{aligned}$$

$$x_1 \in \{1, 2, 4\}$$

$$x_2 \in \{1, 2, 4\}$$

$$x_3 \in \{1, 2, 4\}$$

$$x_4 \in \{\cancel{1}, \cancel{2}, \cancel{4}\}$$


Hall set

n variables whose domains cover exactly **n values**

Hall set

n variables whose domains cover exactly **n values**

$$x_1 \in \{1, 2, 4\}$$

$$H = \{x_1, x_2, x_3\}$$

$$x_2 \in \{1, 2, 4\}$$

$$\bigcup D(x_i) = \{1, 2, 4\}$$

$$x_3 \in \{1, 2, 4\}$$

$$x_4 \in \{\cancel{1}, \cancel{2}, \cancel{4}, 5, 6, 7\}$$

Hall set

n variables whose domains cover exactly **n values**

Every conflict and propagation is explained by a Hall set

Hall set

n variables whose domains cover exactly **n values**

Every conflict and propagation is explained by a Hall set

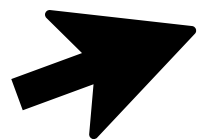


Easy to design a checker!

Valid explanation?



$$\begin{aligned} &\langle x_1 \geq 1 \rangle \wedge \langle x_1 \leq 2 \rangle \wedge \\ &\langle x_2 \geq 1 \rangle \wedge \langle x_2 \leq 2 \rangle \wedge \\ &\langle x_3 \geq 1 \rangle \wedge \langle x_3 \leq 2 \rangle \\ &\Rightarrow \\ &\perp \end{aligned}$$



Reconstruct domains

$$x_1 \in \{1, 2\}$$

$$x_2 \in \{1, 2\}$$

$$x_3 \in \{1, 2\}$$

$$\langle x_1 \geq 1 \rangle \wedge \langle x_1 \leq 2 \rangle \wedge$$

$$\langle x_2 \geq 1 \rangle \wedge \langle x_2 \leq 2 \rangle \wedge$$

$$\langle x_3 \geq 1 \rangle \wedge \langle x_3 \leq 2 \rangle$$

$\Rightarrow$

$\perp$



Reconstruct domains

Check condition

$$\begin{aligned} &\langle x_1 \geq 1 \rangle \wedge \langle x_1 \leq 2 \rangle \wedge \\ &\langle x_2 \geq 1 \rangle \wedge \langle x_2 \leq 2 \rangle \wedge \\ &\langle x_3 \geq 1 \rangle \wedge \langle x_3 \leq 2 \rangle \\ &\Rightarrow \\ &\perp \end{aligned}$$

$$x_1 \in \{1, 2\}$$

$$x_2 \in \{1, 2\}$$

$$x_3 \in \{1, 2\}$$

$$|\{1, 2\}| < |\{x_1, x_2, x_3\}|$$



Explanation verified

Not a valid explanation!



$$\begin{aligned} &\langle x_1 \geq 1 \rangle \wedge \langle x_1 \leq 3 \rangle \wedge \\ &\langle x_2 \geq 1 \rangle \wedge \langle x_2 \leq 2 \rangle \wedge \\ &\langle x_3 \geq 1 \rangle \wedge \langle x_3 \leq 2 \rangle \\ &\Rightarrow \\ &\perp \end{aligned}$$

Reconstruct domains

$$x_1 \in \{1, 2, 3\}$$

$$x_2 \in \{1, 2\} \quad |\{1, 2, 3\}| < |\{x_1, x_2, x_3\}|$$

✗

$$x_3 \in \{1, 2\}$$

Check condition

Explanation rejected

Checker Procedure

1. Reconstruct domains based on the explanation

2. Check condition

$$H = \{x_i\}$$

$$\bigcup D(x_i) < |H| ?$$



Checker Procedure

Easy to verify correctness of explanations!

1. Reconstruct domains based on the explanation

(even without knowing
the conflict/propagation algorithm)

2. Check condition

$$H = \{x_i\}$$

$$\bigcup D(x_i) < |H| ?$$

Another Example: Propagate?

$$x_1 \in \{1, 2\}$$

$$x_2 \in \{1, 2, 3\}$$

$$x_3 \in \{3, 4\}$$

$$x_4 \in \{3, 4, 5\}$$

$$x_5 \in \{4, 5\}$$

$$x_1 \in \{1, 2\}$$

$$x_2 \in \{1, 2, 3\}$$

$$x_3 \in \{3, 4\}$$

$$x_4 \in \{3, 4, 5\}$$

$$x_5 \in \{4, 5\}$$

$$x_1 \in \{1, 2\}$$

$$x_2 \in \{1, 2, \cancel{3}\}$$

$$x_3 \in \{3, 4\}$$

$$x_4 \in \{3, 4, 5\}$$

$$x_5 \in \{4, 5\}$$

Hall set



$$x_1 \in \{1, 2\}$$

$$x_2 \in \{1, 2, \cancel{3}\}$$

$$x_3 \in \{3, 4\}$$

$$x_4 \in \{3, 4, 5\}$$

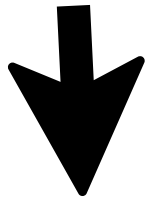
$$x_5 \in \{4, 5\}$$

Hall set

Explanation?



Lift explanation?



$$\langle x_3 \geq 3 \rangle \wedge \langle x_3 \leq 4 \rangle \wedge \\ \langle x_4 \geq 3 \rangle \wedge \langle x_4 \leq 5 \rangle \wedge \\ \langle x_5 \geq 4 \rangle \wedge \langle x_3 \leq 5 \rangle$$

$\Rightarrow$

$$\langle x_2 \neq 3 \rangle$$

$$x_1 \in \{1, 2\}$$

$$x_2 \in \{1, 2, \text{X}\}$$

$$x_3 \in \{3, 4\}$$

$$x_4 \in \{3, 4, 5\}$$

$$x_5 \in \{4, 5\}$$



Lifted explanation

$$\langle x_3 \geq 3 \rangle \wedge \langle x_3 \leq 5 \rangle \wedge \\ \langle x_4 \geq 3 \rangle \wedge \langle x_4 \leq 5 \rangle \wedge \\ \langle x_5 \geq 3 \rangle \wedge \langle x_3 \leq 5 \rangle$$

$\Rightarrow$

$$\langle x_2 \neq 3 \rangle$$

$$x_1 \in \{1, 2\}$$

$$x_2 \in \{1, 2, \cancel{3}\}$$

$$x_3 \in \{3, 4\}$$

$$x_4 \in \{3, 4, 5\}$$

$$x_5 \in \{4, 5\}$$



$$\langle x_3 \geq 3 \rangle \wedge \langle x_3 \leq 5 \rangle \wedge \\ \langle x_4 \geq 3 \rangle \wedge \langle x_4 \leq 5 \rangle \wedge \\ \langle x_5 \geq 3 \rangle \wedge \langle x_3 \leq 5 \rangle$$

$\Rightarrow$

$$\langle x_2 \neq 3 \rangle$$

Convert into conflict explanation

$$\langle x_3 \geq 3 \rangle \wedge \langle x_3 \leq 5 \rangle \wedge \\ \langle x_4 \geq 3 \rangle \wedge \langle x_4 \leq 5 \rangle \wedge \\ \langle x_5 \geq 3 \rangle \wedge \langle x_3 \leq 5 \rangle \wedge \\ \langle x_3 = 3 \rangle$$

$\Rightarrow$

$\perp$

Reconstruct domains

$$\langle x_3 \geq 3 \rangle \wedge \langle x_3 \leq 5 \rangle \wedge \\ \langle x_4 \geq 3 \rangle \wedge \langle x_4 \leq 5 \rangle \wedge \\ \langle x_5 \geq 3 \rangle \wedge \langle x_3 \leq 5 \rangle \wedge \\ \langle x_3 = 3 \rangle$$

$\Rightarrow$

$\perp$

$$x_2 \in \{3\}$$

$$x_3 \in \{3, 4, 5\}$$

$$x_4 \in \{3, 4, 5\}$$

$$x_5 \in \{3, 4, 5\}$$

$$\langle x_3 \geq 3 \rangle \wedge \langle x_3 \leq 5 \rangle \wedge$$

$$\langle x_4 \geq 3 \rangle \wedge \langle x_4 \leq 5 \rangle \wedge$$

$$\langle x_5 \geq 3 \rangle \wedge \langle x_3 \leq 5 \rangle \wedge$$

$$\langle x_3 = 3 \rangle$$

$\Rightarrow$

$\perp$

Reconstruct domains

$$x_2 \in \{3\}$$

$$x_3 \in \{3, 4, 5\}$$

$$x_4 \in \{3, 4, 5\}$$

$$x_5 \in \{3, 4, 5\}$$

Check condition

$$|\{3, 4, 5\}|$$

$<$

$$|\{x_2, x_3, x_4, x_5\}|$$



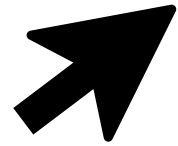
Explanation verified

Propagation for AllDifferent

Compute Hall sets

Prune based on Hall sets

Propagation for AllDifferent



Compute Hall sets

Prune based on Hall sets

All-Different Propagation

variables



x_1

1

x_2

2

x_3

3

x_4

4

All-Different Propagation

Domain values



x_1

1

x_2

2

x_3

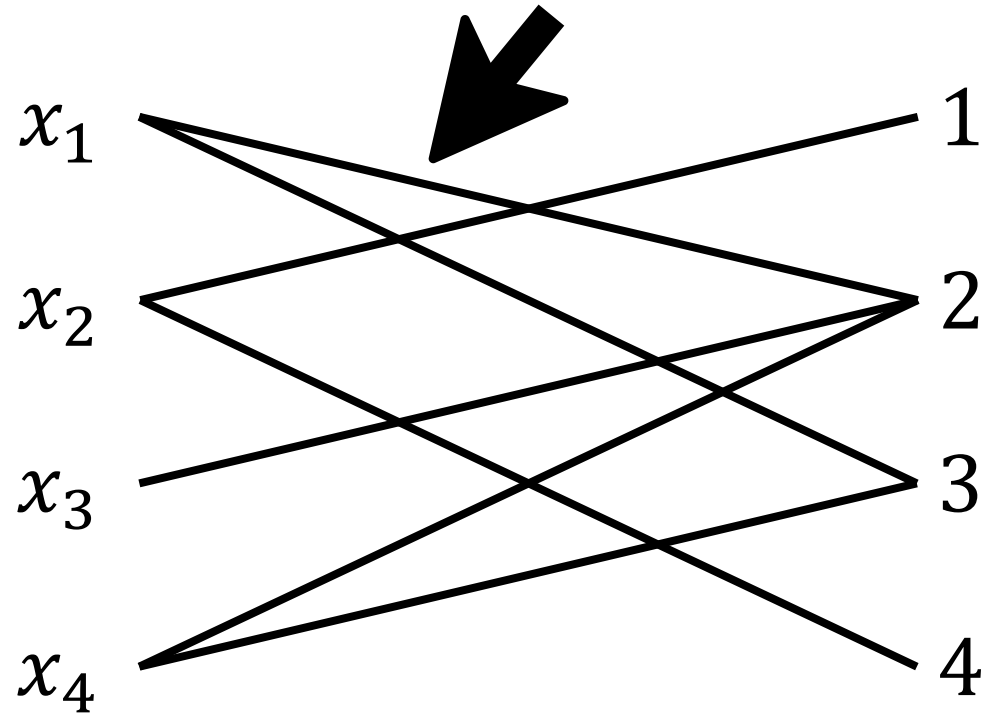
3

x_4

4

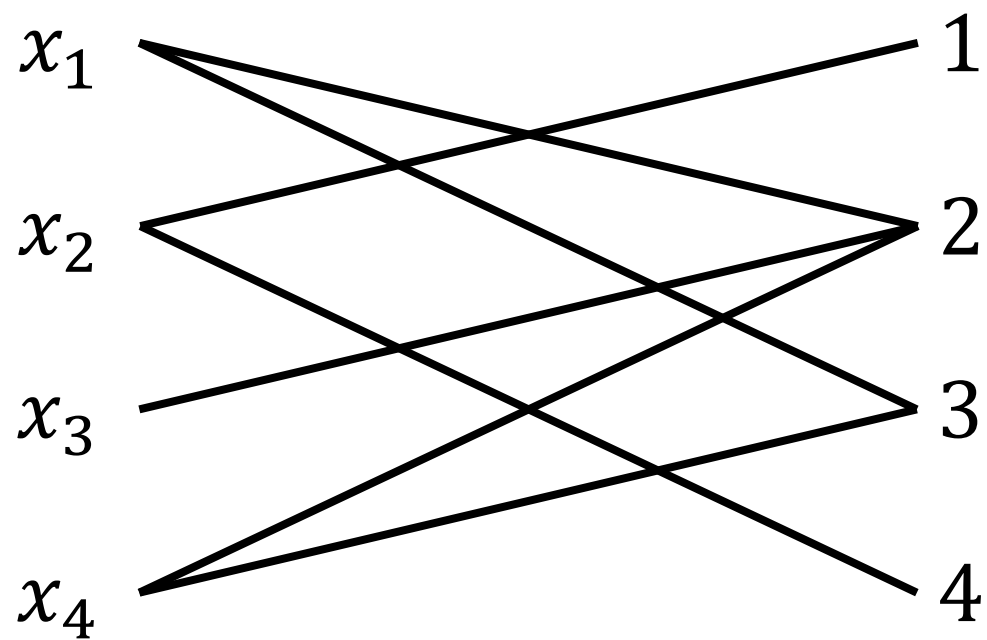
All-Different Propagation

edge connect variable to domain value



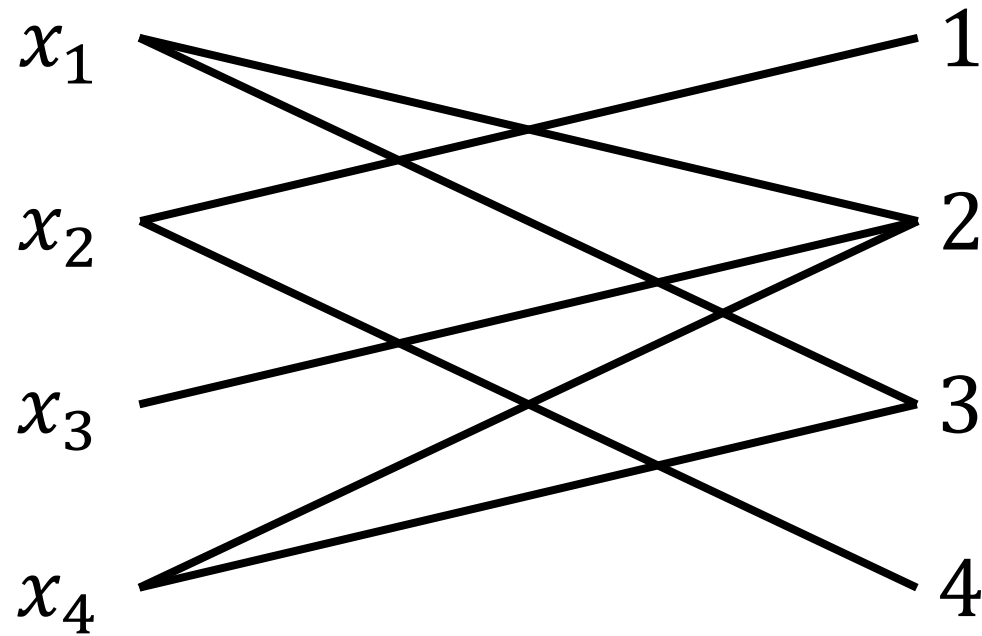
All-Different Propagation

Bi-partite graph



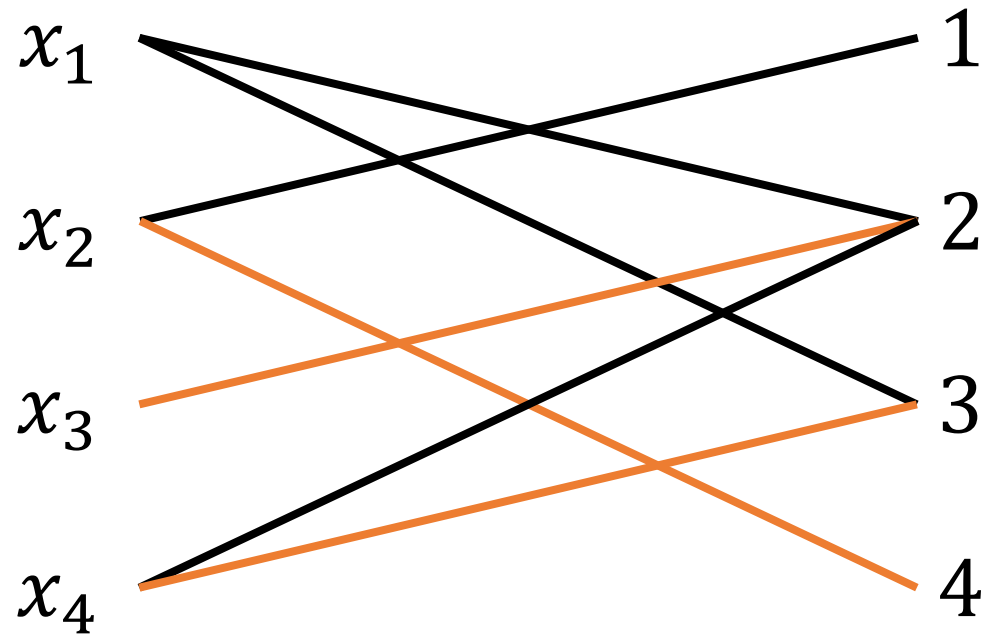
All-Different Propagation

“all-different is feasible if there exists a matching of size four”



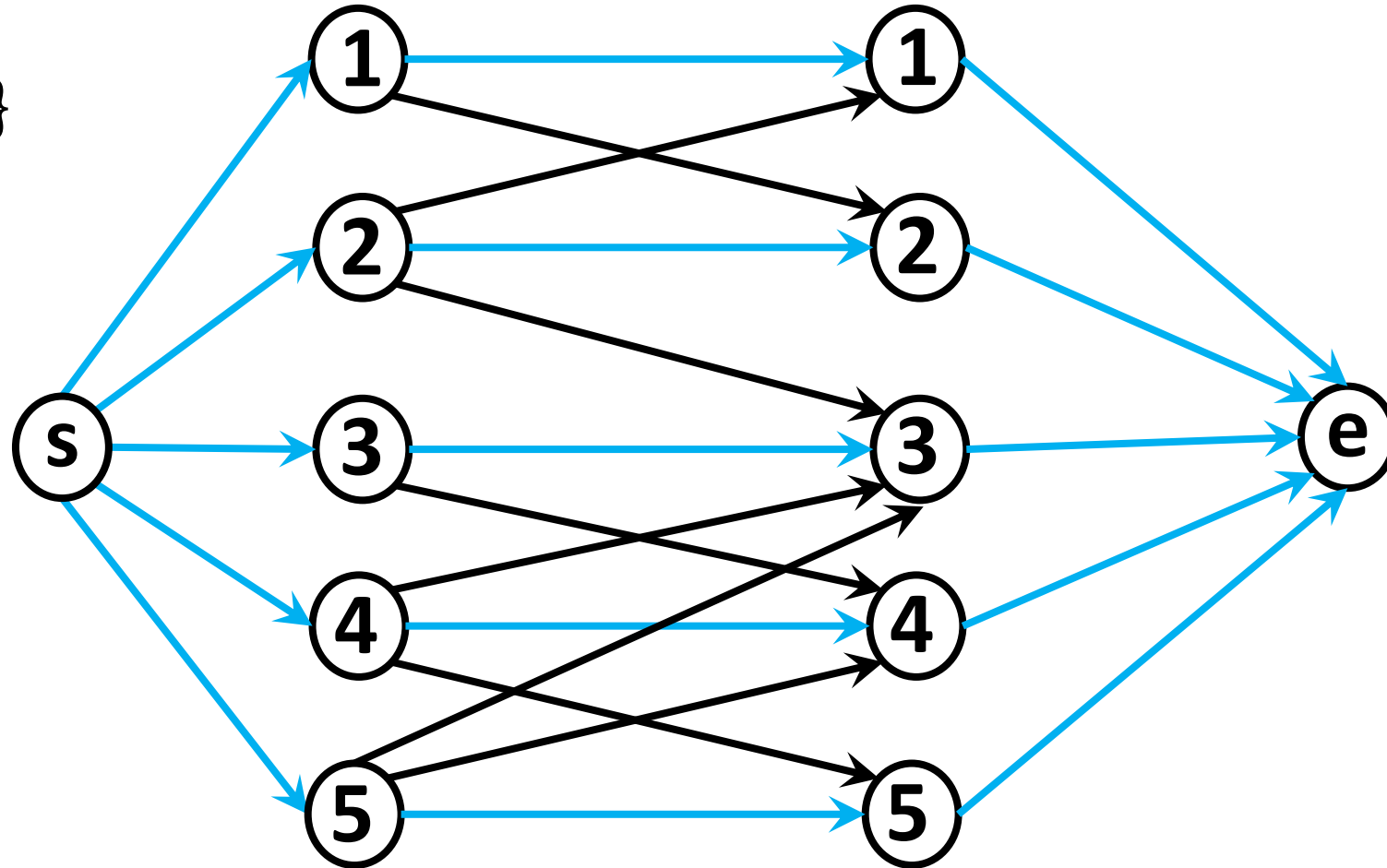
All-Different Propagation

Maximum matching is three, infeasible!



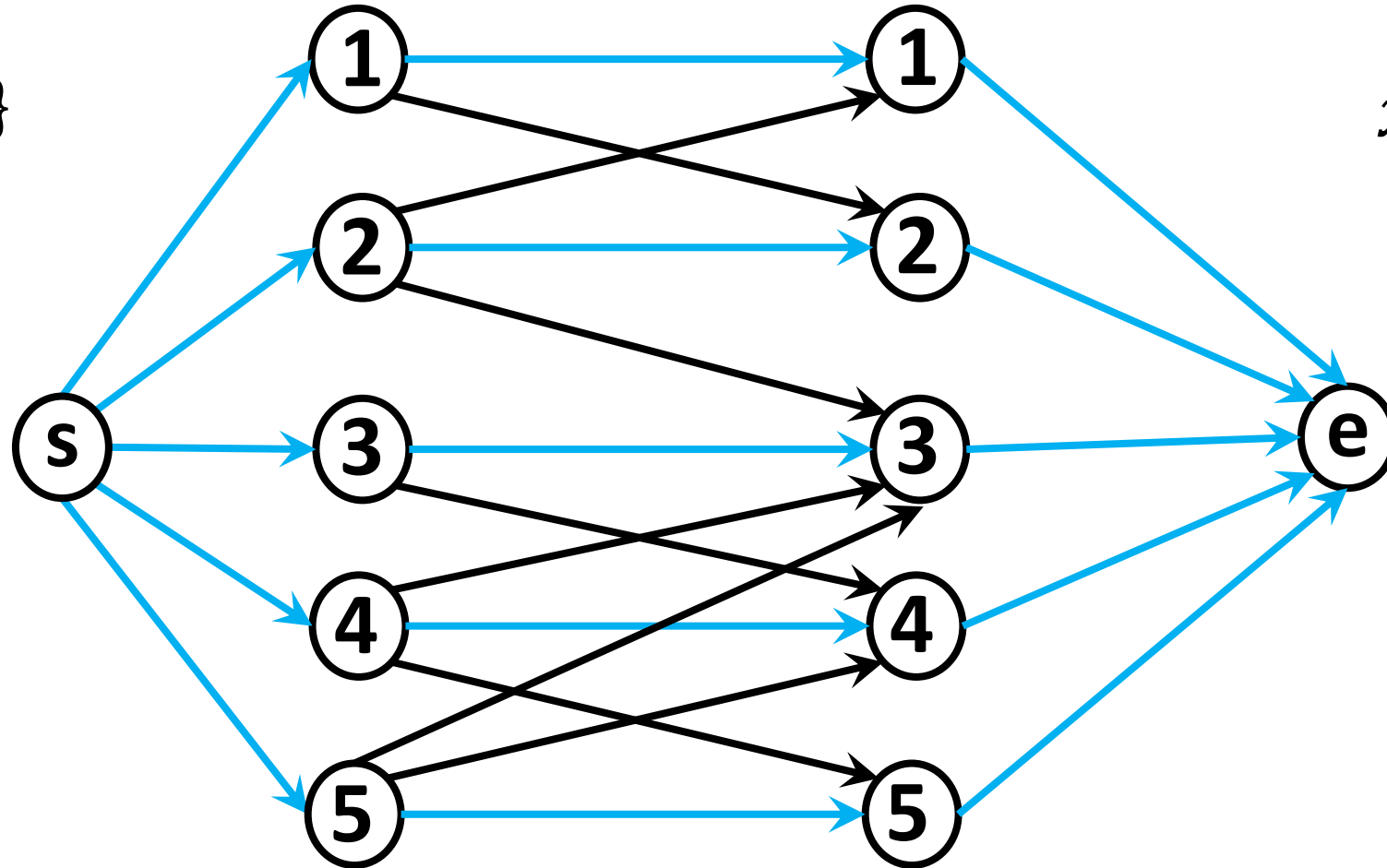
All-Different Flow Representation

$x_1 \in \{1, 2\}$
 $x_2 \in \{1, 2, 3\}$
 $x_3 \in \{3, 4\}$
 $x_4 \in \{3, 4, 5\}$
 $x_5 \in \{4, 5\}$



All-Different Flow Representation

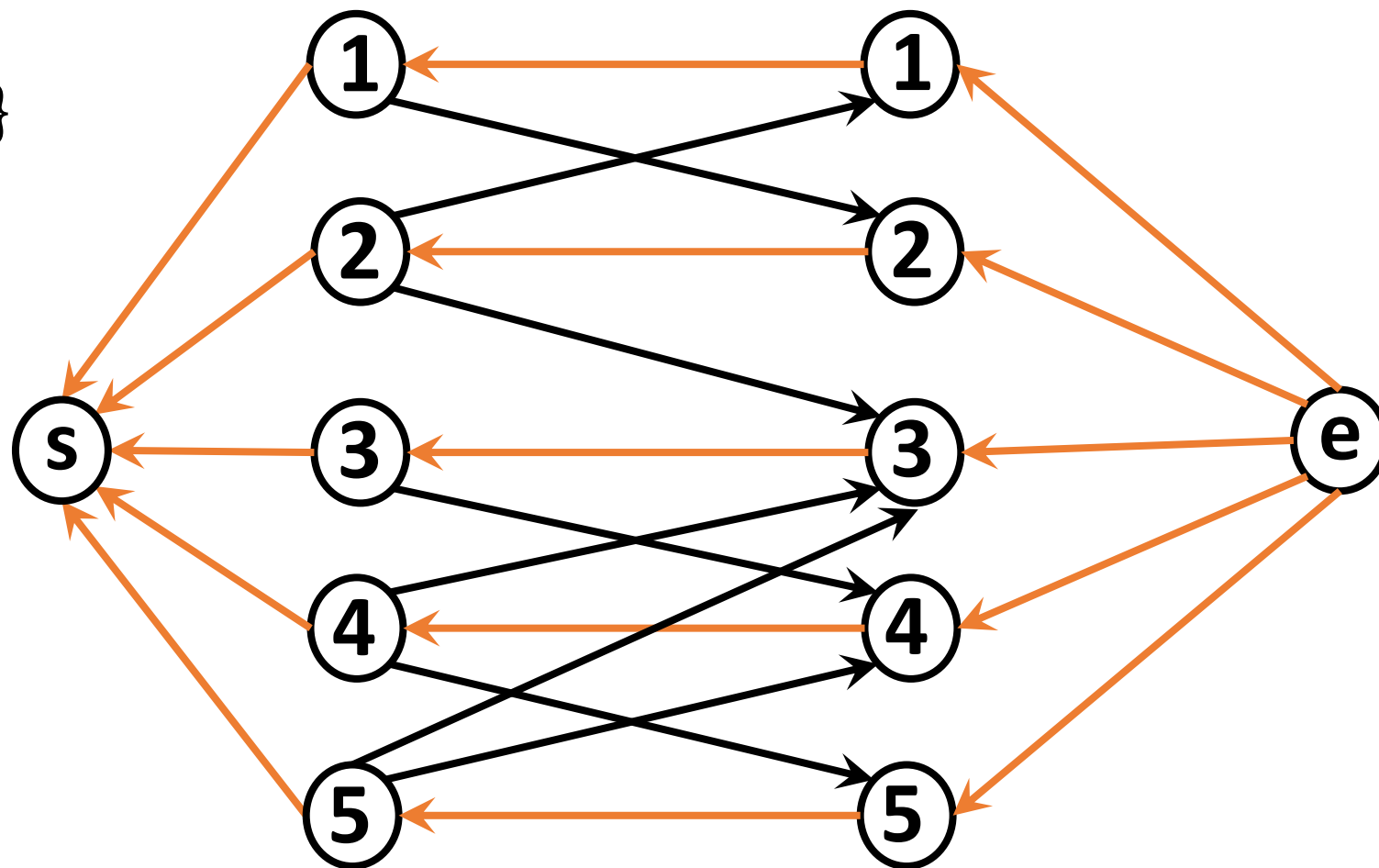
$x_1 \in \{1, 2\}$
 $x_2 \in \{1, 2, 3\}$
 $x_3 \in \{3, 4\}$
 $x_4 \in \{3, 4, 5\}$
 $x_5 \in \{4, 5\}$



$x_i = i$

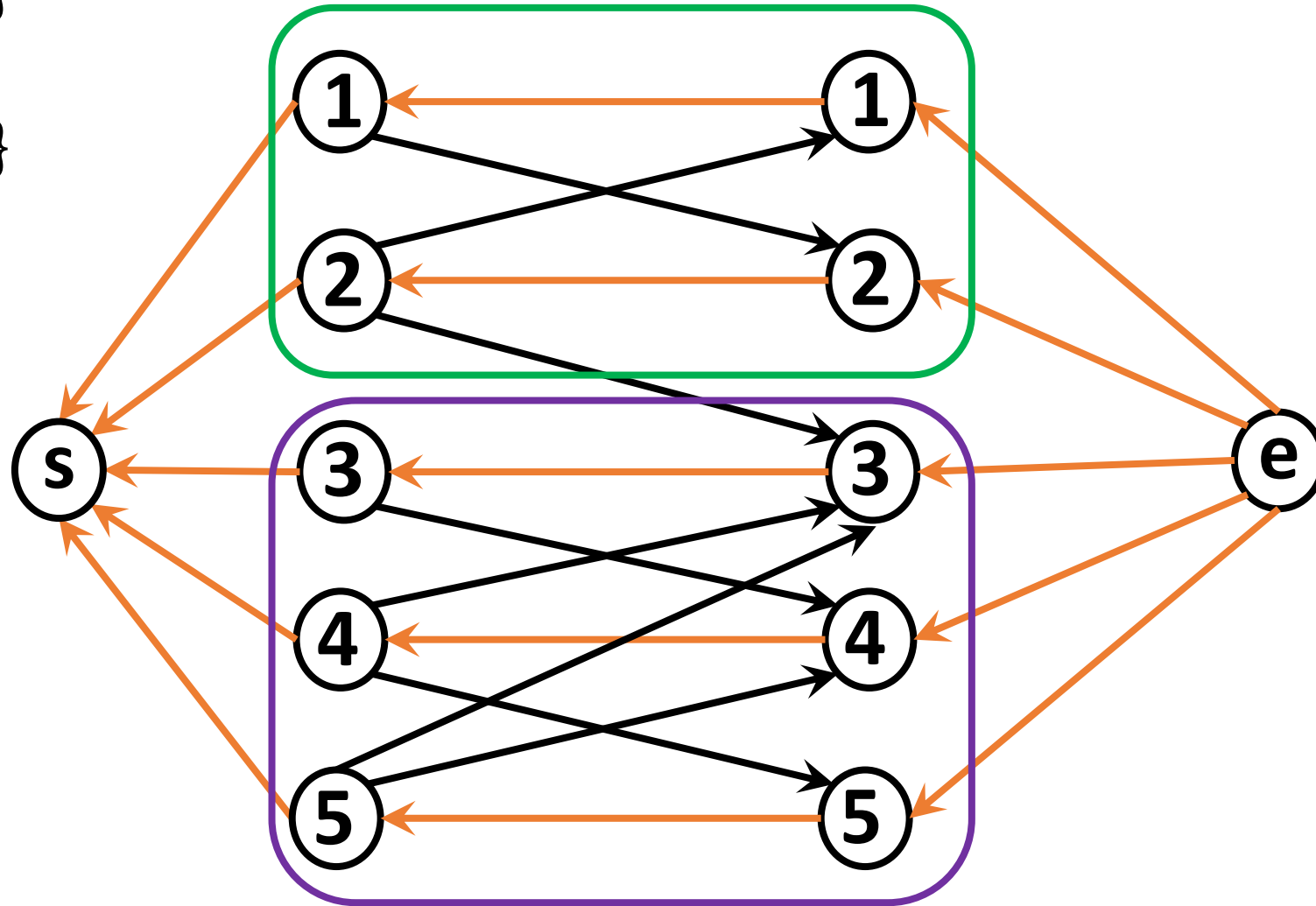
Residual graph: inverse flow edges

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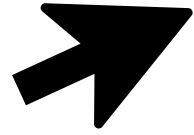


Strongly Connected Components

$x_1 \in \{1, 2\}$
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 $x_3 \in \{3, 4\}$
 $x_4 \in \{3, 4, 5\}$
 $x_5 \in \{4, 5\}$

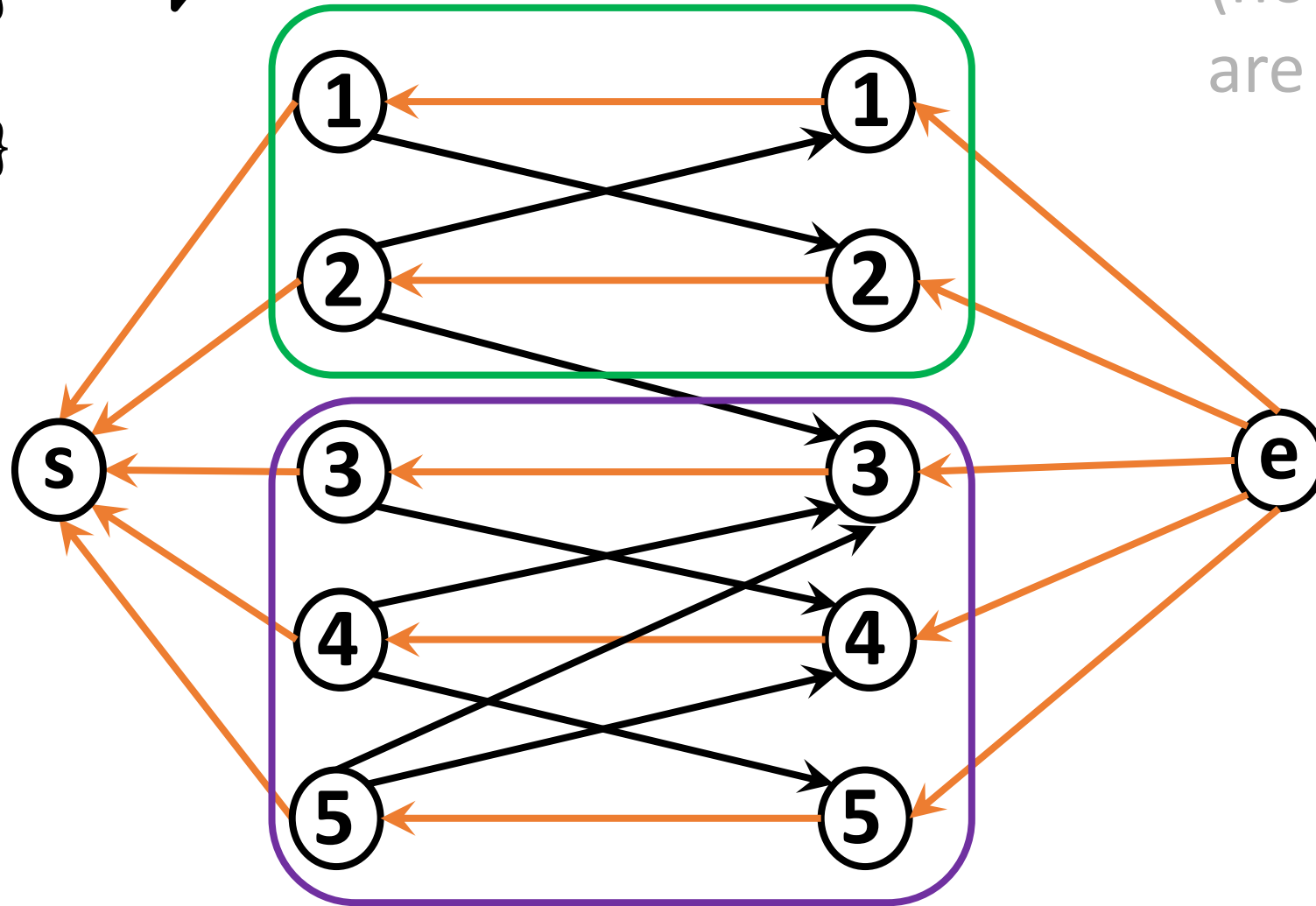


$x_1 \in \{1, 2\}$
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 $x_4 \in \{3, 4, 5\}$
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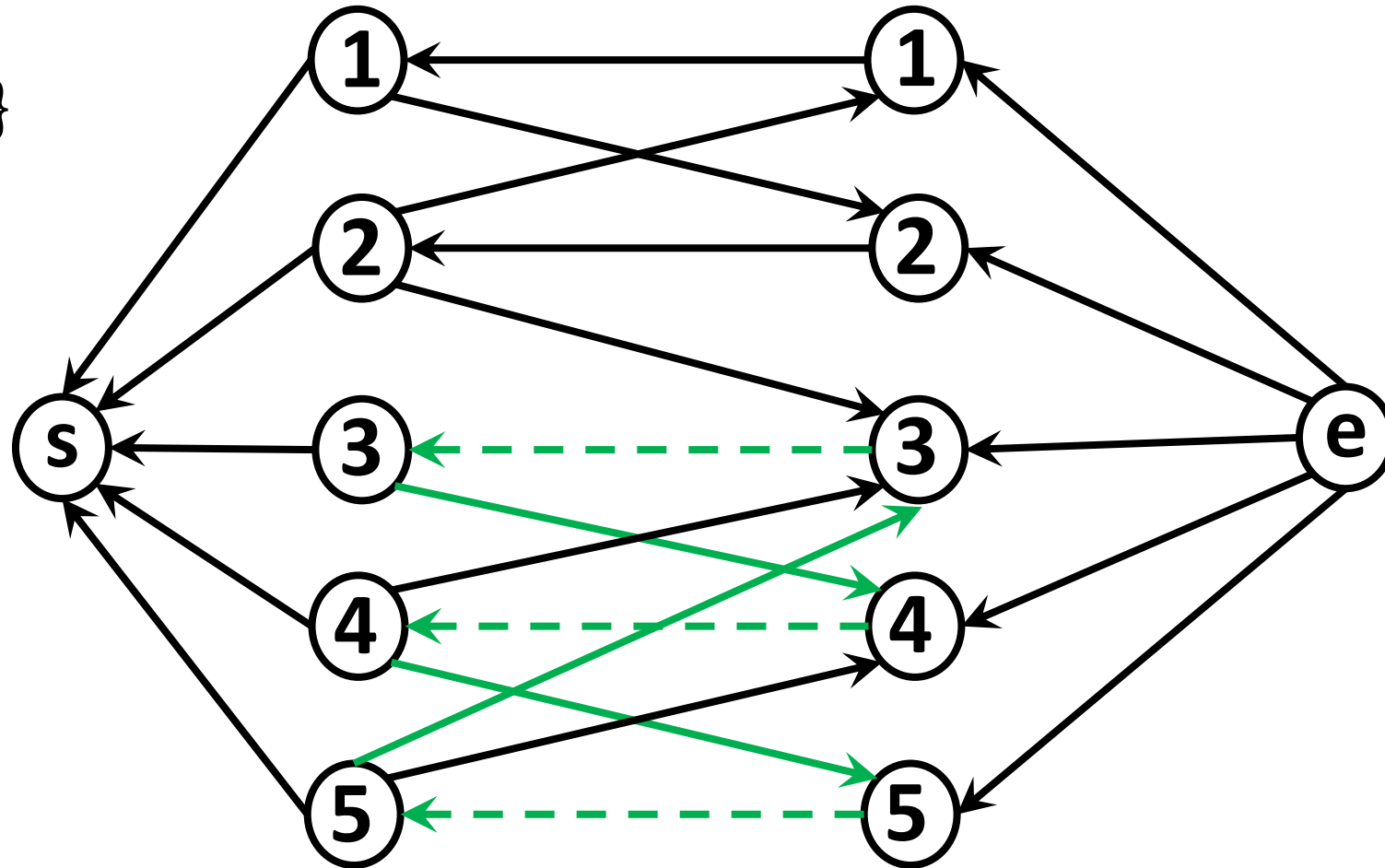
Hall sets form SCCs!

(not all SCCs
are Hall sets)



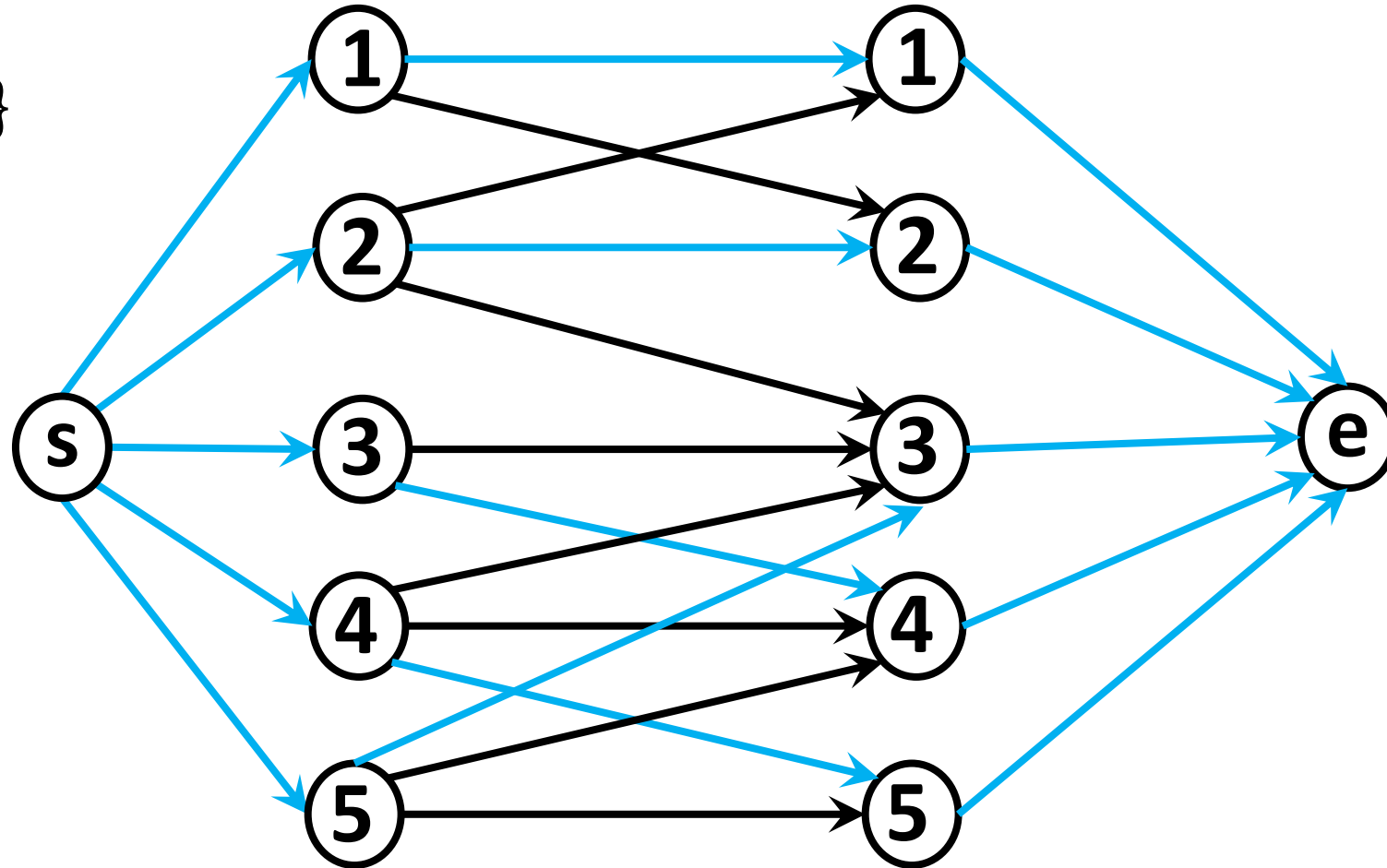
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 $x_3 \in \{3, 4\}$
 $x_4 \in \{3, 4, 5\}$
 $x_5 \in \{4, 5\}$

Cycle within the SCC is a new solution!



$x_1 \in \{1, 2\}$
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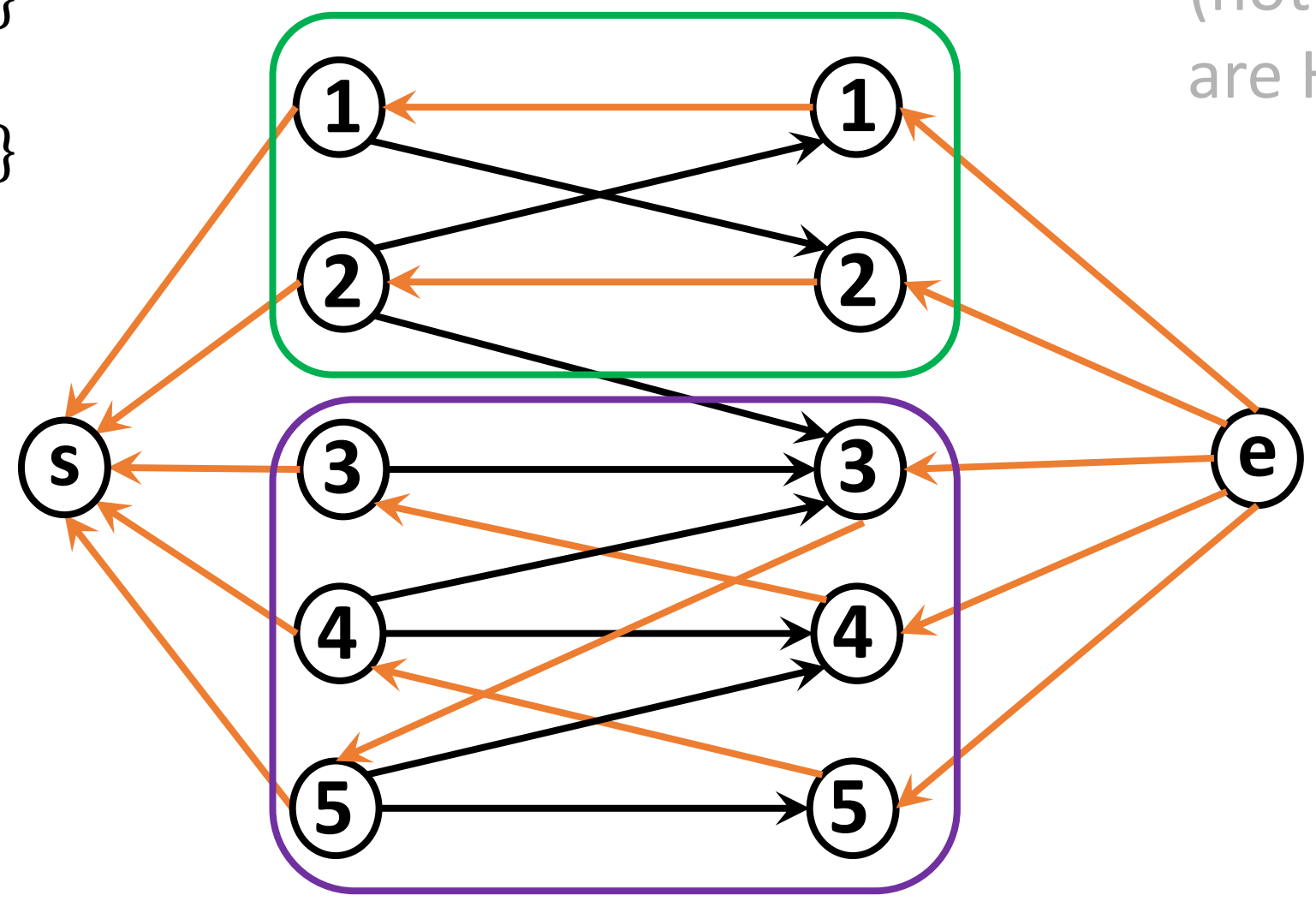
Cycle within the SCC is a new solution!

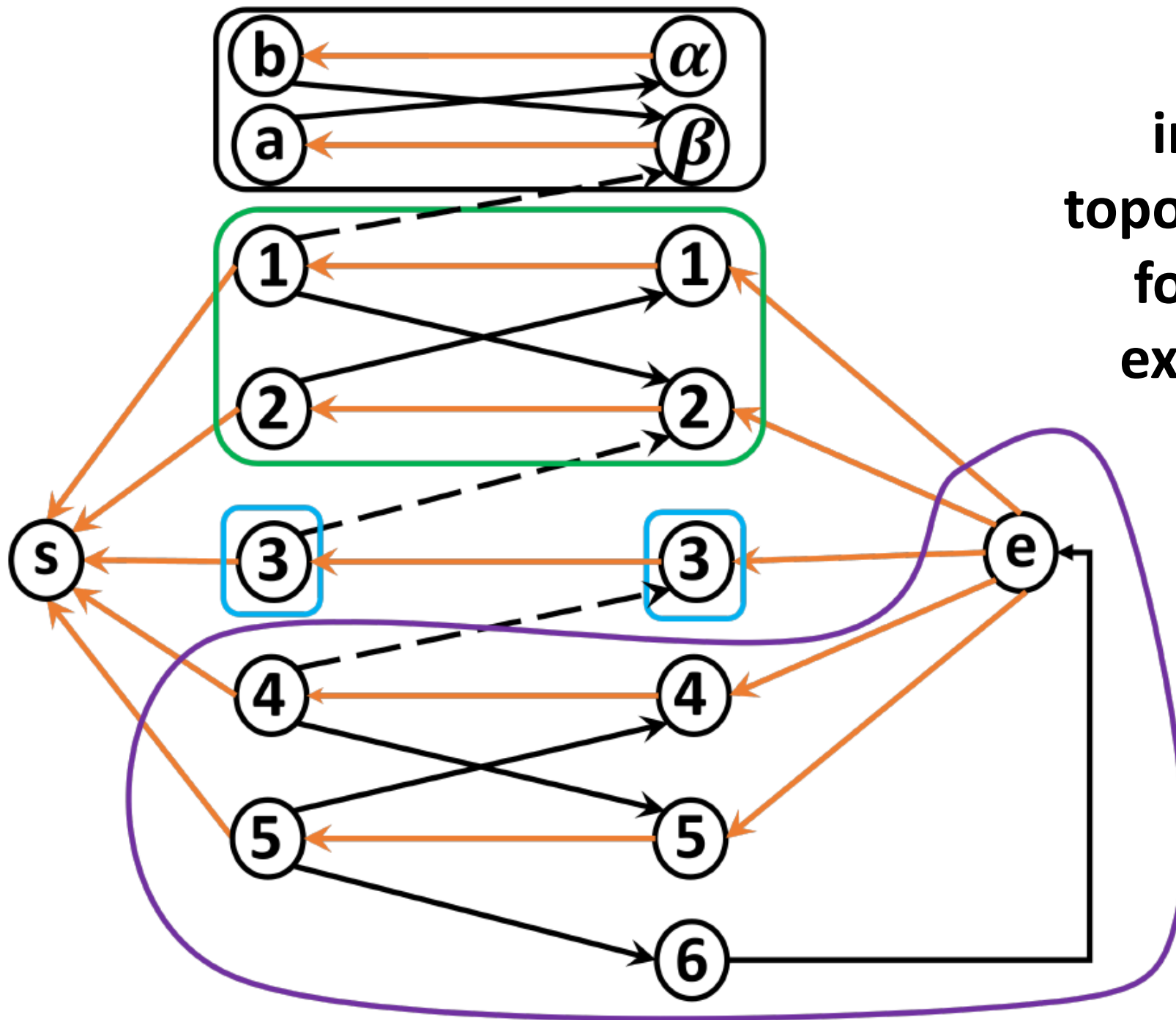


$x_1 \in \{1, 2\}$
 $x_2 \in \{1, 2, 3\}$
 $x_3 \in \{3, 4\}$
 $x_4 \in \{3, 4, 5\}$
 $x_5 \in \{4, 5\}$

Hall sets do not change!

(not all SCCs
are Hall sets)





**Prune
in reverse
topological order
for smallest
explanations**

Summary

Constraint Programming

Propagators and Explanations

Linear inequality: $\sum w_i x_i \geq c$

Max: $x = \max(y, z)$

All-Different

Checkers

Up next...

Conflict Analysis → **Proof-theoretic view of CP**

**Derive new constraints
by combining constraints**

Certificates